THE AI OPERATIONS PLAYBOOK FOR SMALL BUSINESSES

AI Transformation Playbook

AI Adoption Journey



DISCOVER
Identify opportunities and use cases

ADOPT
Engage people and align with business

EXECUTE
Implement and integrate AI into workflows

SCALE
Measure outcomes and scale what works

OUTCOME
Smarter operations. Better decisions. Future ready.

PRACTICAL FRAMEWORK
Step-by-step guidance you can implement

PEOPLE CENTRIC
Designed for adoption and enablement

RESPONSIBLE AI
Governed, secure and ethical

SCALABLE IMPACT
Drive value today. Build advantage tomorrow.

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AI Transformation Playbook

A Practical Framework for AI Adoption, Governance, and Execution — from Boardroom Strategy to Shipped Code

Scope: organizations under 100 employees, designed to scale as headcount grows

Part I: AI Opportunity Discovery • **Part II: AI Adoption Framework** • **Part III: AI Execution Playbook**

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Table of Contents

Executive Summary 12
A Guided Path from an Idea to an Informed AI Decision 12
How the Playbook Is Organized 13
Introduction 14
Getting Your Bearings 14
One Analogy, and the Handful of Terms You Will Need 14
The Running Example: Mylar Technologies 14
How to Use This Playbook 15
Start with the Problem, Not the Technology 15
The Journey This Playbook Follows 16
1. SHOULD we pursue this? 16
2. WHAT should we do first? 16
3. CAN it work? 16
4. CAN we operate it responsibly? 16
5. SHOULD we continue? 16
This Is Not a One-Way Journey 17
When Something Does Not Work 17
The Decisions You Will Make 18
How to Read This Playbook 18
What Success Looks Like 19
Ten AI Transformation Principles 20
1. Measure before you invest 20
2. Chase real pain, not trending use cases 20
3. Govern lightly, but govern first 20
4. A human is always accountable 20
5. Every stage has a diagnostic, not a vibe 20
6. Architecture follows the problem, not the trend 20
7. Credit the reviewer's time, not just the generator's 20
8. Small, reviewed units of change beat large, unreviewed ones 20
9. Distinguish a vendor's claim from an independent finding 20
10. Treat the practice as living, not a launch 21
PART I : AI Opportunity Discovery 22
AI Readiness Assessment 23
Where Most Teams Start 23
A Better Approach 23
Mylar Technologies: Applying This 23
When Things Don’t Go as Planned 24

HEMO - DO NOT COPY -

What You've Achieved 24
Opportunity Identification 25
Setting the Scene 25
Where Most Teams Start 25
A Better Approach 25
Mylar Technologies: Applying This 25
When Things Don’t Go as Planned 26
What You've Achieved 26
Prioritization Matrix 26
Setting the Scene 26
Where Most Teams Start 26
A Better Approach 26
Mylar Technologies: Applying This 26
When Things Don’t Go as Planned 27
What You've Achieved 28
Business Case 28
Setting the Scene 28
Where Most Teams Start 28
Mylar Technologies: Applying This 28
When Things Don’t Go as Planned 29
What You've Achieved 29
AI Architecture Selection 29
Where Most Teams Start 29
A Better Approach 30
Mylar Technologies: Applying This 30
What You've Achieved 33
PART II : AI Adoption Framework 34
Purpose and Scope 35
The Universal AI Adoption Maturity Model 36
The adoption pyramid 37
Minimal roadmap 37
AI Operating Model & Role Responsibility Matrix 38
Level-by-Level Breakdown 38
Chapter 11: Enterprise Level 39
Setting the Scene 39
Where Most Teams Start 39
A Better Approach 39
Foundational Concepts 39

HEMO - DO NOT COPY -

What This Section Helps You Achieve 40
Step by Step — The Five Maturity Stages 41
Stage 1 — Aware 41
Stage 2 — Experimenting 41
Stage 3 — Adopting 42
Stage 4 — Integrated 43
Stage 5 — Optimizing 44
What It Looks Like in Practice 44
Milestone checklist 44
KPIs (quantitative) 44
When Things Don't Go as Planned 45
Scenarios and Use Cases 45
Case study and references 45
Case study: Mylar Technologies (illustrative composite) 45
What You've Achieved 46
Chapter 12: Management Level 47
Setting the Scene 47
Where Most Teams Start 47
A Better Approach 47
Foundational Concepts 47
What This Section Helps You Achieve 47
Step by Step — The Five Maturity Stages 48
Stage 1 — Aware 48
Stage 2 — Experimenting 49
Stage 3 — Adopting 49
Stage 5 — Optimizing 51
What It Looks Like in Practice 51
Milestone checklist 51
KPIs (quantitative) 51
When Things Don't Go as Planned 52
Scenarios and Use Cases 52
Case study and references 52
Case study: Mylar Technologies (illustrative composite) 52
Chapter 13: Team Level 53
Setting the Scene 53
Where Most Teams Start 53
A Better Approach 53
Foundational Concepts 53

HEMO - DO NOT COPY -

What This Section Helps You Achieve 53
Step by Step — The Five Maturity Stages 54
Stage 1 — Aware 54
Stage 2 — Experimenting 55
Stage 3 — Adopting 55
Stage 4 — Integrated 56
Stage 5 — Optimizing 57
What It Looks Like in Practice 57
Milestone checklist 57
KPIs (quantitative) 57
When Things Don’t Go as Planned 58
Scenarios and Use Cases 58
Case study and references 58
Case study: Mylar Technologies (illustrative composite) 58
What You've Achieved 59
Chapter 14: Project Level 60
Setting the Scene 60
Where Most Teams Start 60
A Better Approach 60
Foundational Concepts 60
What This Section Helps You Achieve 60
Step by Step — The Five Maturity Stages 61
Stage 1 — Aware 61
Stage 2 — Experimenting 62
Stage 3 — Adopting 62
Stage 5 — Optimizing 64
What It Looks Like in Practice 64
Milestone checklist 64
KPIs (quantitative) 64
When Things Don’t Go as Planned 65
Scenarios and Use Cases 65
Case study and references 65
Case study: Mylar Technologies (illustrative composite) 65
What You've Achieved 66
Chapter 15: Developer Level 67
Setting the Scene 67
Where Most Teams Start 67
A Better Approach 67

HEMO - DO NOT COPY -

Foundational Concepts 67
What This Section Helps You Achieve 68
Step by Step — The Five Maturity Stages 68
Stage 1 — Aware 68
Stage 2 — Experimenting 69
Stage 3 — Adopting 69
Stage 4 — Integrated 70
Stage 5 — Optimizing 71
What It Looks Like in Practice 71
Milestone checklist 71
KPIs (quantitative) 71
When Things Don't Go as Planned 72
Scenarios and Use Cases 72
Case study and references 72
Case study: Mylar Technologies (illustrative composite) 72
What You've Achieved 73
Chapter 16: QA Level 74
Setting the Scene 74
Where Most Teams Start 74
A Better Approach 74
Foundational Concepts 75
What This Section Helps You Achieve 75
Step by Step — The Five Maturity Stages 75
Stage 1 — Aware 75
Stage 2 — Experimenting 76
Stage 3 — Adopting 76
Stage 4 — Integrated 77
Stage 5 — Optimizing 78
What It Looks Like in Practice 78
Milestone checklist 78
KPIs (quantitative) 78
When Things Don't Go as Planned 79
Scenarios and Use Cases 79
Case study and references 79
Case study: Mylar Technologies (illustrative composite) 79
What You've Achieved 80
Chapter 17: Measuring Success — the "How Do We Know" Scorecard 81
Sample scorecard 81

HEMO - DO NOT COPY -

Illustrative maturity heatmap 82
Chapter 18: Suggested Rollout Timeline 83
Chapter 19: Capability Matrix Across Organization Sizes 84
Chapter 20: Top AI Transformation Mistakes 85
Shadow AI 85
Tool-first thinking 85
No governance 85
Measuring activity instead of value 85
Poor prompt discipline 85
Missing human review 86
Pilot purgatory 86
Confusing a vendor's success story with your own 86
PART III : AI Execution Playbook 87
Chapter 21: Developer Practices — Beyond Basic Adoption 88
Setting the Scene 88
Where Most Teams Start 88
Mylar Technologies: In Practice 89
When Things Don’t Go as Planned 89
What You've Achieved 89
Chapter 22: QA Practices — Beyond Basic Adoption 90
Where Most Teams Start 90
A Better Approach 90
Mylar Technologies: In Practice 91
When Things Don’t Go as Planned 91
What You've Achieved 91
Chapter 23: Agentic AI 92
Setting the Scene 92
Where Most Teams Start 92
A Better Approach 92
Mylar Technologies: In Practice 92
When Things Don’t Go as Planned 94
What You've Achieved 94
Chapter 24: RAG — Retrieval-Augmented Generation 95
Setting the Scene 95
Where Most Teams Start 95
A Better Approach 95
Mylar Technologies: In Practice 96
When Things Don’t Go as Planned 96

HEMO - DO NOT COPY -

What You've Achieved 96
Chapter 25: MCP — Model Context Protocol 97
Setting the Scene 97
Where Most Teams Start 97
A Better Approach 97
Mylar Technologies: In Practice 97
When Things Don't Go as Planned 98
What You've Achieved 98
Chapter 26: Local vs. Cloud Models 99
Setting the Scene 99
Where Most Teams Start 99
Mylar Technologies: In Practice 99
When Things Don't Go as Planned 101
What You've Achieved 101
Chapter 27: ROI Measurement 102
Where Most Teams Start 102
A Better Approach 102
Mylar Technologies: In Practice 103
When Things Don't Go as Planned 103
What You've Achieved 103
Chapter 28: Enterprise Case Studies 104
Setting the Scene 104
Where Most Teams Start 104
Klarna — Agentic AI in production (customer service) 104
LinkedIn — RAG for internal customer-service resolution 104
GitHub Copilot at Accenture — Developer practice transformation 104
Meta — AI-generated testing (QA transformation) 105
DBS Bank — AI governance framework rollout 105
IBM — Reported ROI from internal AI automation 105
When Things Don't Go as Planned 105
What You've Achieved 105
Templates & Worksheets 106
1. AI Readiness Worksheet 106
2. AI Opportunity Canvas 106
3. Business Case Template 106
4. Architecture Decision Record 107
5. Governance Checklist 107
6. Quarterly Review Template 108

HEMO - DO NOT COPY -

7. ROI Calculator 108
8. Risk Register 109
Future Roadmap: AI Evolution 2026–2030 110
2026 — Agent proliferation meets a maturity shakeout, simultaneously 110
2027 — The cancellation wave, and what survives it 110
2028 — Agents as economic actors, not just assistants 110
2029-2030 — Agents, robots, and people become a designed partnership 110
Conclusion & Summary 112
Appendix — Frameworks, Matrices, and Scorecards at a Glance 113
Applying the Framework: Six Standpoints at Mylar Technologies 115
Enterprise Standpoint 115
Management Standpoint 115
Team Standpoint 115
Project Standpoint 115
Developer Standpoint 116
QA Standpoint 116
Glossary 117
Agentic AI 117
AI Readiness Scorecard 117
BYOAI 117
Build vs. Buy vs. Blend 117
Fine-tuning 117
Governance Pillars 117
Hallucination 117
Maturity Ladder / Maturity Stage 117
MCP (Model Context Protocol) 117
Pain Point 117
RACI 117
RAG (Retrieval-Augmented Generation) 117
Risk-Adjusted ROI 118
RICE-A Scoring 118
Shadow AI 118
Test Theater 118
Value Stream Mapping 118
Vendor Case Study vs. Peer-Reviewed Study 118
Workflow 118
References 119
Part I — AI Opportunity Discovery 119

HEMO - DO NOT COPY -

Part II — AI Adoption Framework 119
Part III — AI Execution Playbook 119
Future Roadmap 2026–2030 120

HEMO - DO NOT COPY -

Executive Summary

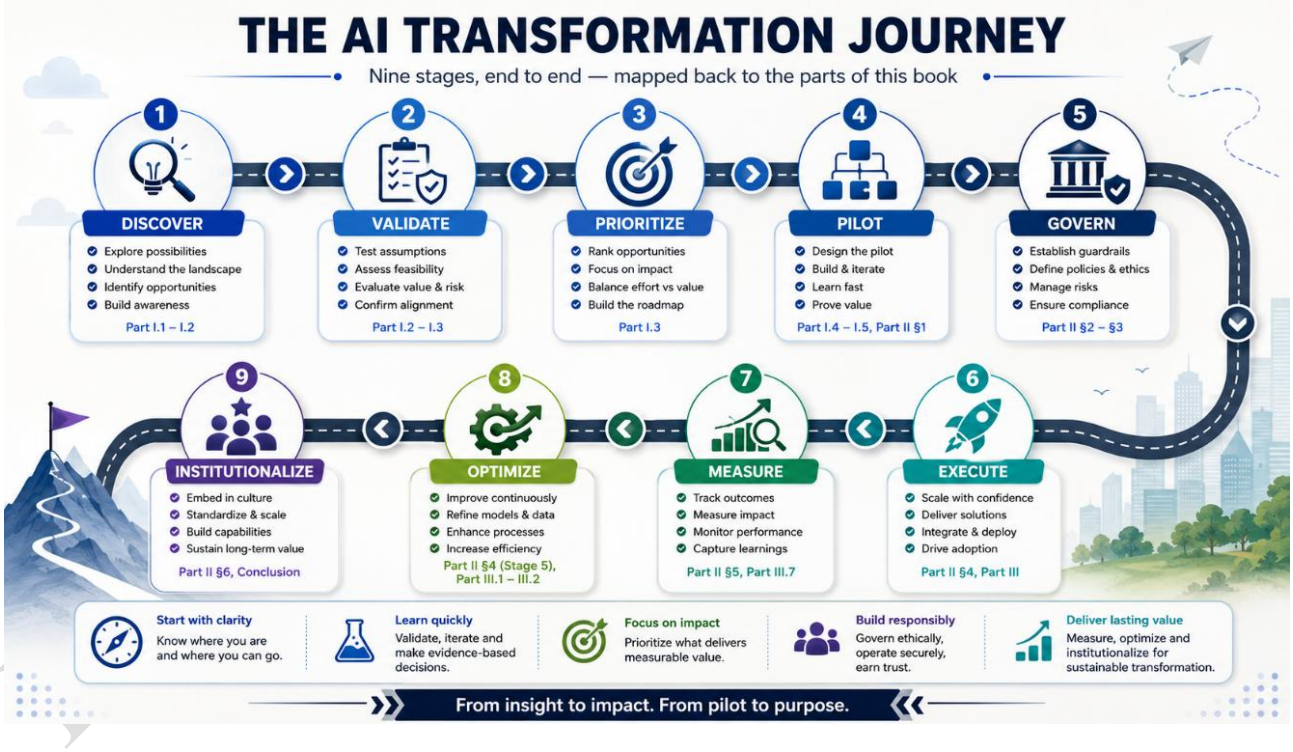
A Guided Path from an Idea to an Informed AI Decision

Mylar Technologies is a small scale software company that builds scheduling and workflow integrations for mid-market clients. For two quarters, its leadership had been talking about AI the way a lot of organizations do: a Slack channel full of links, a few people quietly using ChatGPT on their own, no real decision made. Meanwhile, the support queue was the one place everyone agreed something needed to change — response times had crept from hours to a day as the client base grew, and two new hires were already budgeted just to keep pace.

That is a familiar starting point: an obvious pain point, a fashionable technology, and no clear process connecting the two. Experimentation alone would not have told Mylar's CEO whether an AI tool would actually fix the support queue, whether it was worth the cost, or what to do if the first attempt failed. A tool can produce an impressive response without solving a real business problem. A pilot can look successful without being measurable. A solution can work technically and still fail because the data, workflow, cost, risk, or people around it were not ready.

This playbook follows Mylar's decision from that starting point through to a working, governed AI deployment — and it is written for readers in the same position: new to AI, with a real problem in front of them and no structured way yet to move from instinct to decision. It does not assume the reader already understands AI models, RAG, agents, fine-tuning, governance, or maturity frameworks; these concepts are introduced exactly when they become relevant to a decision Mylar has to make.

The journey begins with a problem, not with a tool.



These are the questions Mylar's leadership had to answer, in order — and the same sequence any reader can apply to their own starting point:

1. What problem are we trying to solve?
2. Is the problem important enough to pursue?
3. Is AI appropriate, or would a simpler solution work better?
4. Do we have the required data, skills, infrastructure, ownership, and support?

5. Which opportunity should we attempt first?
6. What is the smallest useful experiment we can run?
7. How will we measure whether it works?
8. What should we do when the result is inaccurate, expensive, slow, risky, or difficult to adopt?
9. Should we improve the current approach, select another architecture, return to an earlier stage, or stop?
10. Is the solution ready for production?

The playbook organizes this journey into nine connected stages: Discover, Validate, Prioritize, Pilot, Govern, Execute, Measure, Optimize, and Institutionalize.
Each stage explains:

- the problem being addressed
- what the reader needs to do
- what information is required
- what output should be produced
- what success looks like
- what commonly goes wrong
- how to correct the issue
- when to return to an earlier stage
- when to make a Go, Conditional Go, Rework, Alternative, or No-Go decision

The outcome is not automatically an AI implementation. A successful outcome may be proceeding to production, continuing with a limited pilot, changing the use case, choosing a different model or architecture, improving the data or process first, using conventional automation instead of AI, or stopping because the value does not justify the cost or risk.
A No-Go decision is not a failure. It is evidence that the process prevented unnecessary investment.

How the Playbook Is Organized

Part I: AI Opportunity Discovery: Helps the reader understand readiness, find meaningful problems, compare opportunities, build a business case, and choose an initial technical approach.
Part II: AI Adoption Framework: Explains how the organization introduces AI consistently across leadership, management, teams, projects, developers, and QA. It provides maturity stages and diagnostic questions to determine where each area currently stands.
Part III: AI Execution Playbook: Introduces the technical and operational practices required once a use case has been approved, including prompting, RAG, fine-tuning, agents, MCP, testing, monitoring, deployment, governance, and ROI measurement.
The purpose of this book is not to convince every reader to adopt AI. Its purpose is to help readers make better decisions about where AI belongs, where it does not, and how to move from curiosity to a responsible and measurable outcome.

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Introduction

Getting Your Bearings

This book does not assume you have built, bought, or evaluated an AI system before. Background is introduced exactly when it becomes relevant to a decision, not before.

- Every technical term is explained in plain language the first time it is used, and gathered again in the Glossary at the back of the book for quick reference.
- You do not need to read the book cover to cover in one sitting. Each Part stands on its own: start with Part I if you are still deciding whether to pursue AI at all, Part II if your organization already has some AI activity but no consistency across teams, or Part III if a use case has already been approved and you need to build it.
- Chapters follow a consistent pattern — where most teams start, a better approach, what commonly goes wrong, and what you will have achieved — so you always know what is coming next.
- Where a chapter references a framework, template, or tool from elsewhere in the book, a note in the text points you to exactly where to find it.

None of this requires a technical background to follow. Where a decision genuinely requires deeper technical knowledge, the book says so explicitly and points you to the right chapter in Part III for when that time comes.

One Analogy, and the Handful of Terms You Will Need

Think of an AI model the way you would think of a capable new hire on their first day. They arrive with broad knowledge and good judgment from years of general training, but no idea what a particular business needs from them until someone tells them. A handful of terms used throughout this book describe different parts of that same relationship.

- A prompt is the instruction you give them — specific, ambiguous, well-scoped, or vague, exactly the way an instruction to a new hire can be any of those things.
- Retrieval-Augmented Generation (RAG) is handing them the actual customer file before asking a question about it, instead of trusting them to remember it correctly. It is the difference between “what do you know about this account” and “read this account’s file and tell me.”
- Fine-tuning goes deeper than a single instruction. It is specialized training that permanently changes how they approach a whole category of task — closer to a multi-week onboarding program than a single day’s instructions.
- A workflow is a checklist you have designed for them: do step one, then step two, always in that order, regardless of what happens. An agent is closer to handing them the keys and the tools and trusting them to work out the right order themselves for a multistep assignment.
- A hallucination is a confident, fluent, wrong answer — the equivalent of a new hire answering a question without checking, and sounding completely sure while doing it.

Every one of these terms is covered again, in more depth, in the Glossary at the back of the book. Nothing later in this playbook requires memorizing them now — they are introduced again, in context, exactly when each one first matters.

The Running Example: Mylar Technologies

To make the ideas in this book concrete, most chapters return to the same illustrative example: Mylar Technologies, a fictional small scale software company building scheduling and workflow integrations for mid-market clients. The Executive Summary opened with Mylar's starting point — an overloaded support queue and two quarters of unfocused AI talk. The chapters that follow trace what Mylar actually did about it, one decision at a time: what it assessed, what it prioritized, what it built, and what it would do differently.

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Mylar is not a real company, and its story is a composite built to show how one organization might apply this framework at each stage — not a verified case study. Where this book needs a real, published example instead, it says so explicitly and names the actual company (see Chapter 28, Enterprise Case Studies).

How to Use This Playbook

Artificial Intelligence can be surprisingly easy to try and surprisingly difficult to implement well.
A person can open an AI tool, enter a prompt, and receive an impressive response within minutes. A team can build a proof of concept in days. This ease of experimentation creates an understandable temptation to begin with the technology: Which model should we use? Should we build a chatbot? Do we need RAG? Should we create an AI agent? Should we fine-tune a model?
These are useful questions, but they are rarely the best place to start.
The first question should be simpler: What problem are we trying to solve?
This playbook begins there.
It is designed for readers who are new to AI, organizations beginning their AI adoption journey, and teams that may already be experimenting but need a more structured way to move forward.
You do not need to understand every AI technology before using this playbook. The relevant concepts are introduced as they become necessary. The objective is not to turn every reader into an AI expert. The objective is to provide a process for making better decisions about where AI may help, how to test it, what to do when it does not work, and whether it should ultimately proceed to production.

Start with the Problem, Not the Technology

AI is a possible solution. It is not automatically the solution.
A business problem may be solved by:

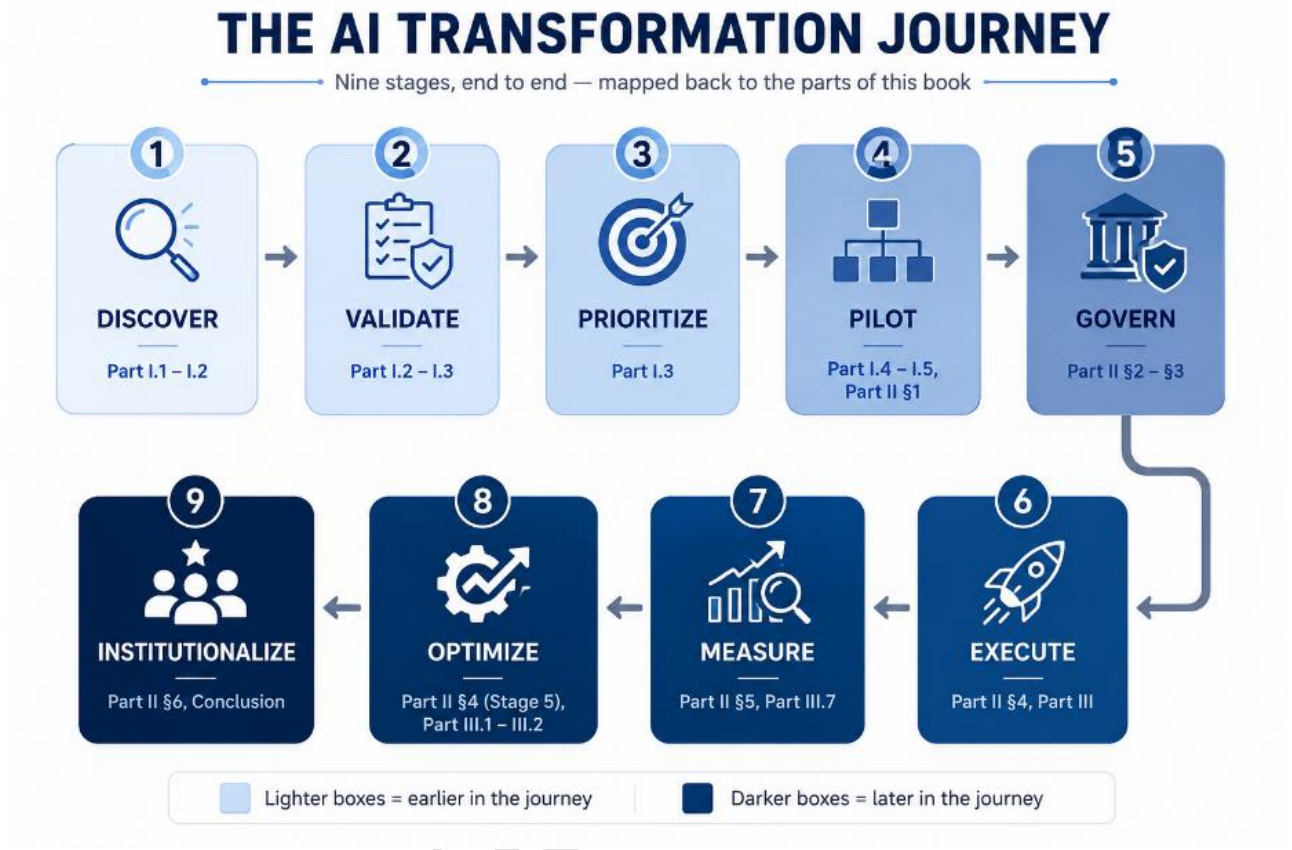
- improving an existing process
- changing a workflow
- using conventional automation
- improving data quality
- integrating existing systems
- applying analytics or rules
- using an AI-assisted approach
- building an AI-enabled solution

The purpose of the early stages of this playbook is to understand the problem before committing to a technology.
For example, consider this statement: “We need an AI chatbot for customer support.” This already assumes the solution.
A better starting point is: “Customers wait too long to receive answers to frequently asked questions, increasing support workload and reducing customer satisfaction.”
Now several solutions can be considered. Perhaps a better knowledge base is enough. Perhaps workflow automation can solve part of the problem. Perhaps an AI assistant can help support staff. Perhaps a customer-facing AI system is appropriate.
When we begin with the technology, we try to find a problem for the solution. When we begin with the problem, we can find the most appropriate solution.

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The Journey This Playbook Follows

The playbook guides an idea through a structured journey:
Discover → Validate → Prioritize → Pilot → Govern → Execute → Measure → Optimize → Institutionalize



1. SHOULD we pursue this?

First, understand the problem. Is it real? Who experiences it? How significant is it? Is there evidence? Is solving it valuable? Is AI appropriate? This is the purpose of Discover and Validate.

2. WHAT should we do first?

Most organizations will identify more opportunities than they can pursue. Prioritize the opportunity with the best combination of value, feasibility, readiness, and manageable risk.

3. CAN it work?

The purpose of a Pilot is not to prove that AI is impressive. It is to test specific assumptions about quality, data, adoption, performance, cost, and risk. A successful pilot produces evidence. An unsuccessful pilot also produces evidence.

4. CAN we operate it responsibly?

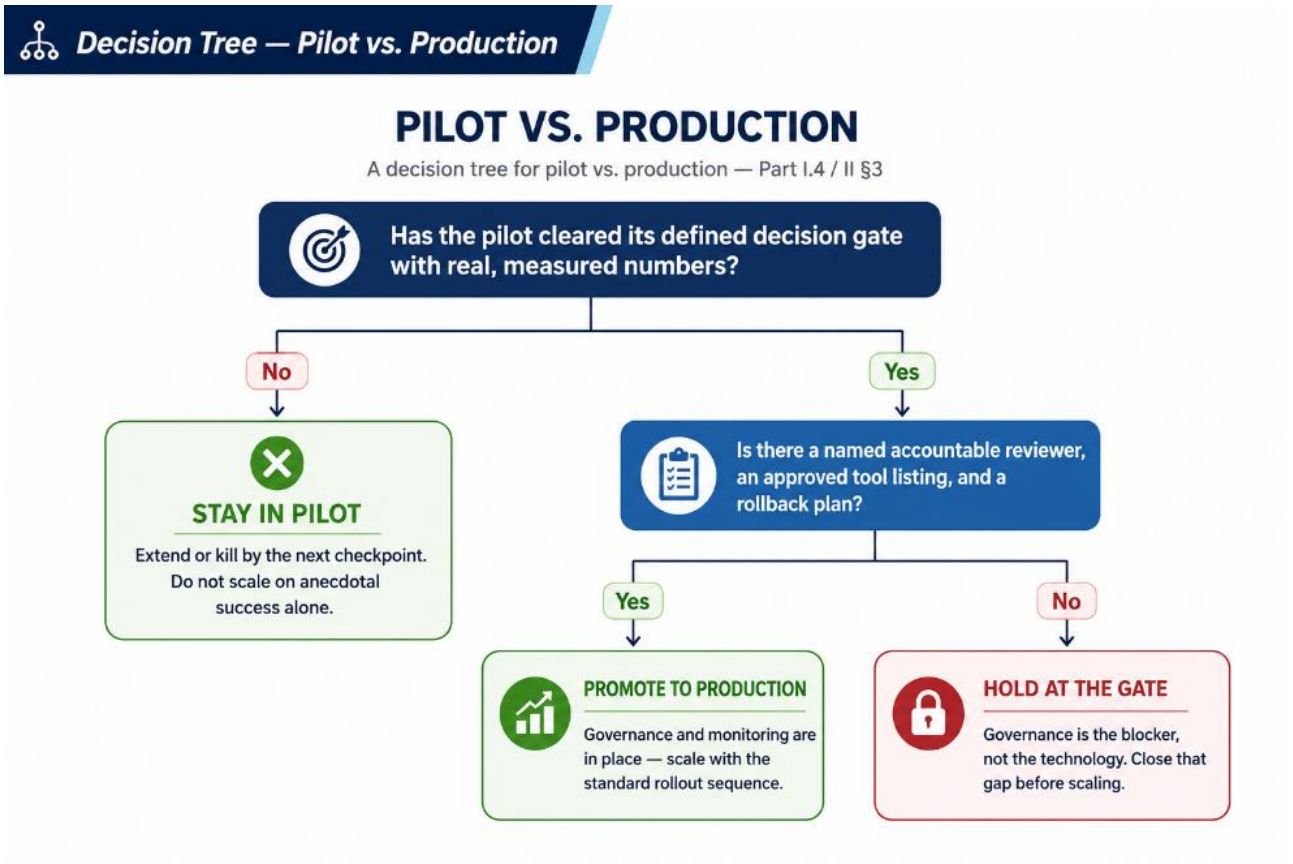
A demonstration and a production system are very different things. Govern, Execute, and Measure determine whether the solution can operate reliably, securely, economically, and responsibly.

5. SHOULD we continue?

Once a solution is operating, continue asking whether it delivers value, whether quality and adoption are improving, whether costs or risks have changed, and whether the solution should be expanded, redesigned, replaced, or retired.

This Is Not a One-Way Journey

One of the most important ideas in this playbook is that progress does not always mean moving to the next stage. A result may require you to move backward. A pilot may reveal poor data. Poor data may require additional preparation. A technically successful solution may have poor user adoption. A solution with excellent quality may be too expensive. An AI approach may turn out to be unnecessary because conventional automation can solve the problem more reliably and economically.



Returning to an earlier stage is not failure. Changing the approach is not failure. Making a No-Go decision is not failure.

The purpose of the process is not to force every idea into production. The purpose is to reach the best decision with the available evidence.

When Something Does Not Work

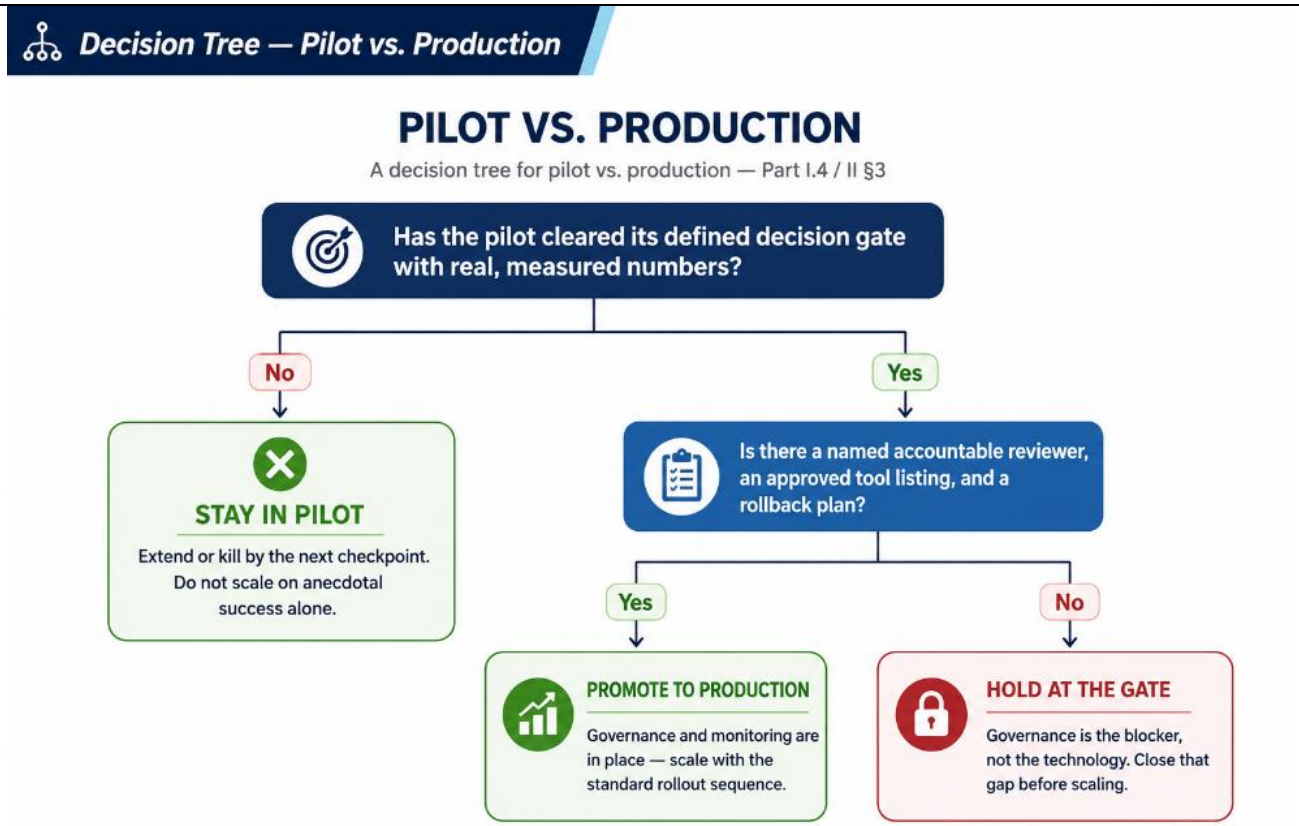
AI projects can fail for many different reasons. The natural reaction to a poor result is often: "We need a better model." Sometimes that is correct. Often it is not.

Before changing the model, diagnose the problem.

- **Was the problem poorly defined?** Return to problem discovery and redefine the use case.
- **Is the required information missing or unreliable?** Improve the data or knowledge source before continuing.
- **Is the prompt or context insufficient?** Improve the prompt, context, examples, or workflow.
- **Is the selected AI approach unsuitable?** Evaluate an alternative technical approach.
- **Is the model insufficient?** Evaluate another model after confirming that the model is actually the constraint.
- **Does the solution work technically but fail operationally?** Redesign the operating model, architecture, workflow, or deployment approach.
- **Is AI simply the wrong solution?** Choose the simpler solution.

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The Decisions You Will Make



Go: The available evidence supports moving to the next stage.
Conditional Go: The opportunity remains viable, but specific gaps must be addressed before proceeding.
Rework: The underlying opportunity remains valid, but the current implementation or assumption needs to change.
Alternative: The problem is worth solving, but the current solution is not the best approach.
No-Go: The evidence does not justify continuing. Capture what was learned so the same experiment is not repeated without new evidence.

How to Read This Playbook

You may read the book from beginning to end, but it is also designed to be used as a working guide.

- If you are just beginning your AI journey, start with Opportunity Discovery.
- If you already have several AI experiments but lack consistency, use the Adoption Framework to assess gaps.
- If you already have an approved use case and need to build or scale it, use the Execution Playbook.

At each stage, ask:

- What problem are we addressing?
- What evidence do we currently have?
- What assumption are we testing?
- What result do we expect?
- What happened?
- Why did it happen?
- What should we do next?

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What Success Looks Like

The success of this playbook should not be measured by the number of AI projects launched. It should be measured by the quality of the decisions made.

- **Production:** The solution demonstrates sufficient value, feasibility, quality, safety, and operational readiness.
- **Continued experimentation:** The opportunity remains promising, but more evidence is required.
- **Redesign:** The problem is valid, but the current approach needs to change.
- **Alternative solution:** A different AI or non-AI approach is more appropriate.
- **Foundation building:** The organization needs to improve data, skills, infrastructure, governance, or processes before continuing.
- **No-Go:** The evidence shows that the initiative should not proceed.
- All of these can be valid outcomes.

The objective is not simply, "How do we implement AI?"
The better question is: "How do we determine where AI creates value, test that assumption with evidence, respond intelligently when it does not work, and make the right decision about what to do next?"

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Ten AI Transformation Principles

Frameworks are followed; principles are remembered. The ten statements below compress everything in this book into something a team can recall in a hallway conversation, the way engineering teams recall the Well-Architected pillars or the Agile Manifesto's four values. Each one is a direct, practical lesson from a specific part of this book, not an abstract aspiration.

1. Measure before you invest

Run the readiness assessment (Chapter 2) before the tool purchase, not after. An honest score that says "not ready" is worth more than an optimistic guess that says "ready."

2. Chase real pain, not trending use cases

Opportunity identification (Chapter 3) exists because the AI opportunity that gets funded should be the one solving a documented bottleneck, not the one that was easiest to imagine.

3. Govern lightly, but govern first

A one-page policy and a named accountable owner (Chapter 10) take days to put in place and prevent the incidents that take months to clean up. Governance is a pace-setter, not a bureaucratic tax.

4. A human is always accountable

AI is never the accountable party for a decision (Chapter 10). Every AI-assisted output has a named human who owns whether it ships.

5. Every stage has a diagnostic, not a vibe

"Which stage are we at" is answered with the five diagnostic questions per stage (Chapters 11–16), not with whoever in the room has the strongest opinion.

6. Architecture follows the problem, not the trend

Prompting, RAG, and fine-tuning solve different problems (Chapter 6, Chapter 24). Choosing based on which one is most talked about is how expensive rework gets built in from day one.

7. Credit the reviewer's time, not just the generator's

The single biggest ROI measurement error (Chapter 27) is crediting an AI with the two minutes it took to generate an output while ignoring the ten minutes a human spent reviewing and correcting it.

8. Small, reviewed units of change beat large, unreviewed ones

AI-authored pull requests run roughly 51% larger than human ones by default, which erodes review quality (Chapter 21). Keep the unit of change small enough that a human reviewer can actually hold it in their head.

9. Distinguish a vendor's claim from an independent finding

Every case study in this book (Chapter 28) is labeled by evidentiary strength for a reason: a company's marketing about its own success and a peer-reviewed independent study are not the same kind of evidence, even when both are true.

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10. Treat the practice as living, not a launch

Adoption that survives the founding team's turnover is documented, taught in onboarding, and revisited on a quarterly cadence (Chapter 17, Conclusion). A transformation that depends on one champion isn't institutionalized yet — it's a personality.

CHAPTER RECAP

1. Ten memorable principles compress this book's lessons, the way the Agile Manifesto or the AWS Well-Architected pillars do for their own disciplines.
2. Each principle cross-references the specific chapter it's drawn from — use them as a working checklist, not a slogan on a wall.

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PART I : AI Opportunity Discovery

Before adopting AI at scale, know whether you're ready, find where it actually helps, decide what to do first, build the case leadership can approve, and choose the right technical approach. This part turns that discovery work into five repeatable steps.

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AI Readiness Assessment

Where Most Teams Start

Most organizations skip straight to picking a use case or buying a tool, treating a champion's enthusiasm and conviction as sufficient justification. The result: pilots stall not because the model failed, but because the underlying data was inaccessible, no one owned governance, or leadership support evaporated once budget scrutiny began. Readiness is rarely checked until after the first project has already struggled.

A Better Approach

Rather than guessing, or defaulting to whichever use case someone is excited about, the better approach is to score the organization itself first — honestly, and across the dimensions that actually determine whether AI takes root. That means treating readiness as a measurable precondition, not an afterthought, before any tool spend or pilot commitment is made.

Before spending on tools or running a pilot, it pays to know honestly where the organization actually stands. A readiness assessment scores the organization across the dimensions that determine whether an AI initiative takes root — not just the technology, but leadership commitment, data quality, talent, and process maturity around it.

Mylar Technologies: Applying This

Mylar's CEO ran the six-dimension scorecard after two quarters of unfocused AI talk. The result was uneven: Leadership & Strategy and Talent & Skills each scored a 2 — there was interest but no named owner and no one had used an AI tool for real work yet. Data Readiness scored a 4 — support tickets and account history were already clean and centralized in one system. Governance scored a 1 — no policy existed at all. The total, 14 out of 30, placed Mylar in the middle tier: ready for a single, tightly scoped pilot, not a broad rollout. The CEO's first reaction was disappointment; the more useful reaction, and the one the framework asks for, was to treat 14 as a baseline rather than a verdict — the first honest data point the company had about itself.

> *The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.*

Goal

A completed, honest one-page scorecard that tells you whether you're ready for a single focused pilot, a multi-area rollout, or need to build foundations first — before any money is committed.

This adapts two well-documented enterprise models — Gartner's AI Maturity Model (7 pillars: Strategy, Use-Case Portfolio, Governance, Engineering, Data, Ecosystems, People/Culture) and Accenture's "Art of AI Maturity" scorecard (17 capabilities across 4 categories, scored 0-100) — down to six practical dimensions sized for a small organization. Score each 1 (not present) to 5 (strong):

1. **Leadership & strategy:** Is there executive sponsorship and a clearly stated AI ambition, not just enthusiasm?
2. **Data readiness:** Is the data relevant to your candidate use cases accessible, reasonably clean, and governed?
3. **Talent & skills:** Do people have, or are being actively given, the skills to use AI tools well?
4. **Infrastructure:** Do you have the compute, tooling, and integration capacity the use case needs?
5. **Process maturity:** Are your workflows documented well enough to know exactly where AI would plug in?
6. **Governance:** Is there a lightweight AI policy and a named accountable owner (see Chapter 10)?

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Scoring

Sum the six scores (maximum 30). Under 12: not ready for anything beyond a single tightly scoped pilot — spend the next quarter on foundations. 12–21: ready for a focused rollout in one or two areas. 22 or above: ready for a multi-area program across several of the six levels in Part II.



Output

A completed six-dimension scorecard with a stated readiness tier. This becomes the direct input to opportunity identification — you now know how much appetite and capacity you actually have.

> *"Trust is one of the differentiators between success and failure for an AI or GenAI initiative."*
> — Birgi Tamersoy, Gartner (2025)
>
> *"Achievers — the top-performing quartile — reach a median maturity score of 64 out of 100; the largest group, Experimenters, sits at a median of 29."*
> — Accenture, "The Art of AI Maturity" (2026 report, paraphrased from the published quartile data)

When Things Don't Go as Planned

If you can't score a dimension honestly — say, no one can characterize data quality, or no executive will state an AI ambition — treat that gap itself as the finding: score it low rather than skipping it, and flag it as the first foundation to fix. A low overall score isn't a failure; it's a signal to spend a quarter on fundamentals before revisiting the assessment.

What You've Achieved

With the scorecard complete, you have more than a number — a defensible, dimension-by-dimension picture of exactly where the organization is strong and where it isn't. That readiness tier now becomes the input to the next step: identifying which opportunities are realistic to pursue, at what scale, and what still needs to be built first.

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Opportunity Identification

Setting the Scene

This chapter picks up right after readiness: you've decided AI is worth pursuing, and now the question shifts from "should we?" to "where, specifically?" That question matters because most AI budget is wasted not on bad models but on solving the wrong problem — a problem nobody in the building actually has.

Where Most Teams Start

Most teams start from the tool, not the problem: someone saw a demo, a vendor pitch, or a competitor's press release, and now wants to "do something with AI." Whatever use case is trending gets picked, and a pain point is retrofitted afterward to justify it — rather than the pain point driving the choice. No scoring, no process view, just enthusiasm looking for a home.

A Better Approach

The better approach flips the sequence: find the pain first, in a real process, then ask whether AI is the right fix for it. That requires structured evaluation rather than gut feel — which is what the tools in this chapter provide.

Once you know you're ready — or ready enough to start — the next step is finding where AI actually helps. The discipline here is locating real, specific pain, not chasing whatever use case is trending.

Mylar Technologies: Applying This

With a readiness tier in hand but no named opportunity yet, Mylar's Head of Support mapped the actual ticket-resolution workflow rather than starting from a tool. The pain point wasn't vague: response times had crept from hours to a day over two quarters of client growth, and two new support hires were already budgeted just to keep the queue moving. Value stream mapping showed most of that time went to searching account history and prior tickets for context the rep already had access to, not to the actual writing of a reply. That gap — not “we should use AI for support” — became the documented opportunity.

> *The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.*

Goal

A short list of candidate opportunities, each grounded in a real process or pain point rather than a hypothetical.

Three complementary methods, useful in combination:

1. **AI Opportunity Canvas:** A structured, roughly 45-minute worksheet run per candidate process: problem statement, current process, data available, AI capability match, expected value, risks, and a go/no-go call. Run it with the person who does the work plus one technical evaluator.
2. **Pain point × pressure matrix:** Before canvasing everything, score each candidate pain point on a Pain Scale (1–10, how much it hurts) and a Pressure Gauge (1–10, how urgent or visible it is). Only the highest-scoring pain points earn a full canvas.
3. **Value stream mapping:** For a more rigorous, process-level view: map the actual flow of work end to end to see where waste and delay accumulate — that's where an AI-assisted step has the best odds of mattering.

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Output

A list of candidate opportunities, each with a completed (or at least started) AI Opportunity Canvas, ready to be ranked.

> *"Without value-stream mapping (VSM), you can't see where waste accumulates or where AI addresses measurable bottlenecks."*
> — Steve Pereira, Lean.org (2026)
>
> *"Process mining surfaced a loan-processing workflow that ran 35 minutes end to end; after redesign it ran in 5."*
> — Piraeus Bank case, via Celonis methodology (as reported 2025-2026)

When Things Don't Go as Planned

If a full value stream map isn't feasible yet — the process owner's time or the data isn't available — don't let that stall the whole exercise. Run the pain x pressure matrix and a canvas on your top candidates now, and flag the process for VSM once access opens up; a partially completed canvas is still enough to support a go/no-go call.

What You've Achieved

With that list in hand, the guesswork is gone: every candidate has been checked against a real pain point and scored, not just pitched. The next step is ranking them against each other, which is where the following chapter picks up.

Prioritization Matrix

Setting the Scene

By the time a team has run opportunity discovery, it usually has more AI ideas than it has capacity, budget, or credibility to pursue. That surplus is a good problem to have, but it creates a new risk: without a disciplined way to choose, teams default to funding whatever is loudest or newest, and momentum from discovery gets wasted on the wrong bets.

Where Most Teams Start

Most teams prioritize AI opportunities the same way they prioritize everything else: gut feel, seniority of the requester, or whichever idea got the most enthusiastic reaction in a meeting. It's fast, but it's not repeatable, and it rarely holds up when someone asks why one project got funded over another.

A Better Approach

With a list of candidate opportunities in hand, decide what to do first. Not everything valuable is feasible right now, and not everything feasible is valuable enough to bother with. The fix is to score every candidate against the same fixed set of dimensions, so the ranking comes from the criteria rather than from whoever is in the room.

With a list of candidate opportunities in hand, decide what to do first. Not everything valuable is feasible right now, and not everything feasible is valuable enough to bother with.

Mylar Technologies: Applying This

The support opportunity wasn't the only candidate. Engineering had floated AI-assisted code review, and QA wanted automated test generation. Scored side by side with RICE-A, support triage won on Reach (every ticket touches it) and Confidence (the pain point was directly observed, not assumed), even though Effort was moderate and the AI Ambiguity discount was real — nobody at Mylar had built a retrieval-based tool before. Code review and test generation stayed on the list for later, once the first pilot had established a working pattern to reuse.

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The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.

Goal

A ranked shortlist — realistically the top one to three opportunities — that becomes the input to a business case.

A five-dimension scoring framework, each scored 1 (weak), 3 (moderate), or 5 (strong):

1. **Business impact:** How much does solving this actually move a metric leadership cares about?
2. **Technical feasibility:** A score of 5 means straightforward integration with no infrastructure change; 1 means it requires new infrastructure or unproven capability.
3. **Data readiness:** Is the data this needs already accessible and usable, or does it need to be built first?
4. **Strategic alignment:** Does this connect to where the business is actually trying to go?
5. **Speed to value:** Could this show a real result in weeks, or does it only pay off after a long build?



Output

Plot every candidate on a simple Value vs. Effort quadrant (see the infographic below) — Quick Wins get built first, Strategic Bets get a business case, Minor Automations get scheduled opportunistically, and Distractions get shelved. Teams wanting a single composite number can use RICE-A — the standard RICE score (Reach × Impact × Confidence ÷ Effort) with an added "AI Ambiguity" factor that discounts for how unproven the specific AI capability still is.

When Things Don’t Go as Planned

Sometimes every candidate scores low on data readiness, or the team can't agree on a feasibility score because no one has validated the underlying capability yet. When that happens, don't force a ranking on guesses — treat the scoring gap itself as the next task (a short technical spike or data audit) and revisit the matrix once real answers replace assumptions.

What You've Achieved

With scores assigned and candidates plotted, you're no longer holding a pile of ideas — you have a short, defensible list of the one to three opportunities worth taking to a business case, and a documented reason for why the rest can wait.

Business Case

Setting the Scene

By this point in Part I, you've identified and prioritized an AI opportunity — but prioritization alone doesn't authorize spend. Before a team commits engineering time, compute budget, and leadership attention, someone has to write down what the initiative is expected to cost, what it's expected to return, and what could go wrong. Skipping that step is how promising pilots quietly become permanent budget lines that no one can say are actually working.

Where Most Teams Start

Most teams reach for a slide deck: a rough ROI estimate, a demo, and an appeal to competitive pressure. Build cost gets estimated; run cost — the ongoing compute, maintenance, and oversight — usually doesn't. Risk gets a bullet point, if it appears at all, and there's rarely a stated point at which the initiative would be stopped or changed rather than left to continue on momentum.

Before committing real budget, translate the prioritized opportunity into a business case leadership can approve or reject with eyes open — not a pitch, a decision document.

Mylar Technologies: Applying This

The business case compared the AI-assisted approach against the two already-budgeted support hires. The pilot's projected cost, including a modest infrastructure spend and the time to build and review it, was well under one hire's first-year cost, with a faster payback if response time dropped even modestly. The risk-adjusted case discounted the raw time-savings estimate by an assumed error rate, since a wrong answer pulled from the wrong account record would cost more in trust than the time it saved, and the case still cleared the bar. That discounted number, not the optimistic one, is what the CEO actually approved.

> *The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.*

Goal

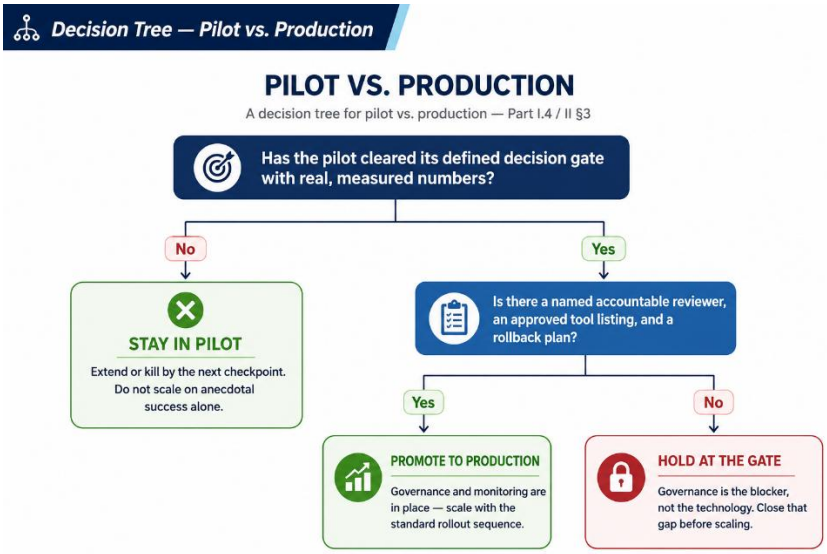
A one-to-two page business case per prioritized opportunity, sufficient for a real go/no-go decision at a defined checkpoint.

Structure the benefits case using McKinsey's five-layer measurement framework, and the risk section using NIST's AI risk taxonomy:

1. **Financial impact:** The bottom-line number — cost saved, revenue enabled, or cost avoided.
2. **Strategic outcomes:** What this does for competitive position, not just this quarter's budget.
3. **Operational KPIs:** The concrete metrics from Part II's scorecard this initiative is meant to move.
4. **Adoption & engagement:** Will people actually use this, and how will you know?
5. **Technical performance:** Accuracy, latency, and reliability thresholds the solution has to clear.

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Decision Tree — Pilot vs. Production



Output

A completed business case covering: problem statement, expected benefit (mapped to the five layers above), total cost (build and run, not just build), a risk assessment drawing on NIST AI 600-1's risk categories most relevant here (hallucination, bias, privacy, vendor lock-in are the ones that come up most often), a payback-period estimate, and a defined decision gate — a specific checkpoint where the initiative is continued, adjusted, or killed, not left to run indefinitely on momentum.

> *"More than 95% of respondents expect moderate to significant value increases in the coming year."*
> — Deloitte, 2025 Tech Value Survey — cited as a confidence-vs-reality gap worth scrutinizing, not taking at face value

When Things Don't Go as Planned

If you can't get a reliable cost or benefit number — pricing isn't finalized, or the KPI baseline doesn't exist yet — say so explicitly in the business case rather than forcing a false-precision estimate, and flag it as an assumption to revisit at the decision gate. If the risk assessment surfaces something blocking, the business case should recommend a delay or a scoped pilot instead of full production, not paper over it to keep the initiative moving.

What You've Achieved

With this in hand, you have more than an idea leadership might like — you have a decision document: a bounded cost and benefit estimate, a named risk profile, and a checkpoint at which the initiative gets continued, adjusted, or killed. That's what turns a prioritized opportunity into something an organization can actually commit real budget to.

AI Architecture Selection

Where Most Teams Start

Left unmanaged, this decision usually happens by default rather than by design: whoever builds the first prototype picks a model because it's familiar or trending, RAG or fine-tuning gets bolted on later when something breaks, and

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build-vs-buy is treated as one blanket choice instead of three separate ones. Nobody writes down why — so when a vendor changes pricing or a new engineer joins, the reasoning has to be reconstructed from memory.

A Better Approach

The shift is to treat architecture selection as three linked but distinct decisions — build vs. buy, model/provider, and prompting vs. RAG vs. fine-tuning — each made deliberately, with its reasoning captured in writing rather than left implicit. That written reasoning is what lets leadership sanity-check a technical choice without needing to be technical themselves, and what makes the decision cheap to revisit later instead of expensive to unwind.

Once a business case is approved, decide how to actually build it. This is a technical decision, but one leadership needs to understand well enough to sanity-check — the wrong architecture choice here is expensive to unwind later.

Mylar Technologies: Applying This

The architecture decision came down to prompting, fine-tuning, or RAG. Fine-tuning was set aside first: Mylar's account data changes daily, and a fine-tuned model would need constant retraining to stay current. Prompting alone was set aside next: a generic answer without the client's actual account history was exactly the trust risk the readiness assessment had flagged. RAG fit the actual requirement — retrieve the specific account's ticket history and documentation at the moment a rep asks, rather than relying on a model's memory or a static instruction. That decision, made here, is the one Part III revisits in technical detail once the pilot is ready to build.

> *The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.*

Goal

A one-page architecture decision record: which build/buy mode, which model or provider and why, and which of prompting, RAG, or fine-tuning fits the use case — with the reasoning written down, not just the conclusion.

Three linked decisions:

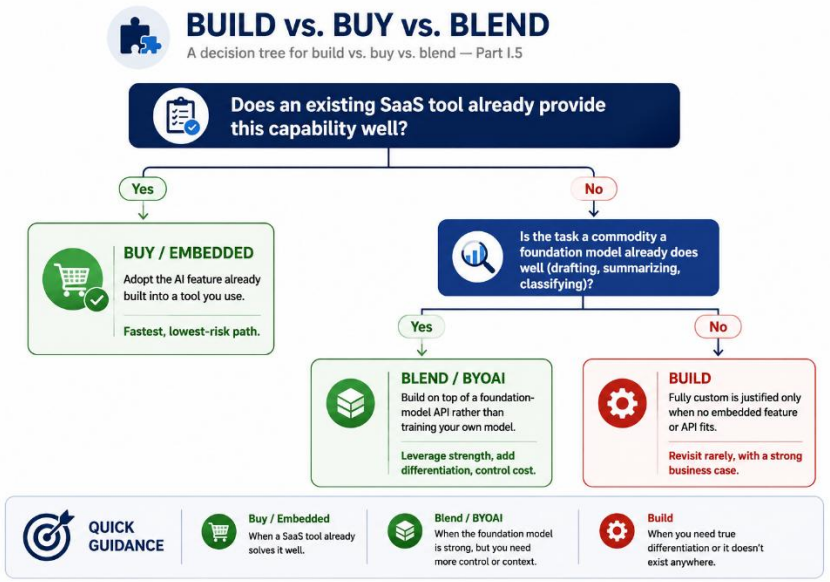
1. **Build vs. buy:** Gartner frames this as three modes, not a strict binary: Embedded (AI already baked into a SaaS tool you use), BYOAI (building on top of a foundation-model API), and Built-and-Blended (a mix). Most organizations this size should default to Embedded or BYOAI rather than a fully custom build.
2. **Model / provider selection:** Choose based on capability, cost, latency, context window, and data-handling terms. Microsoft's own architecture guidance is explicit that the publisher's brand should not be a decision factor on its own — the criteria should be.
3. **Fine-tune vs. RAG vs. prompting:** A simple diagnostic: if the problem is that the model lacks current or proprietary context, that's a RAG problem. If the problem is that the model's style or behavior needs to be reliably consistent, that's a fine-tuning problem. Most organizations should start with prompting and RAG, and only fine-tune once volume and stability justify the investment — this mirrors a standard maturity ladder from prompting, to RAG, to full pipelines, to fine-tuning.

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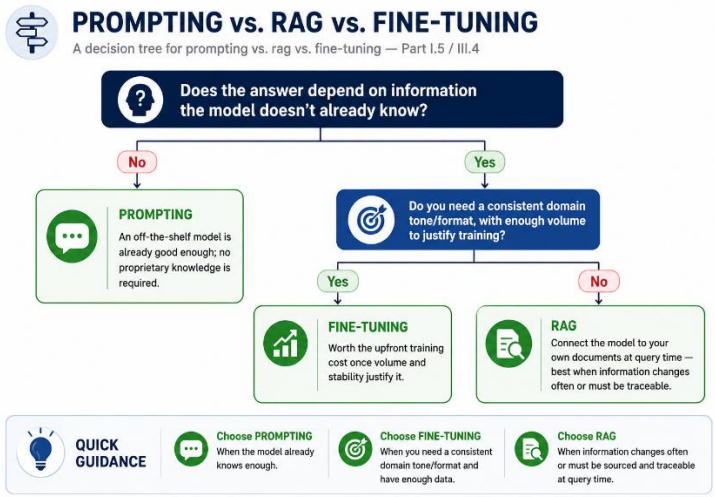
Decision Tree — Model / Provider Selection



Decision Tree — Build vs. Buy vs. Blend



Decision Tree — Prompting vs. RAG vs. Fine-Tuning



Output

A short architecture decision record capturing the build/buy mode chosen, the model/provider and why, and the prompting/RAG/fine-tuning choice with its rationale — durable enough that a new hire could read it and understand why the system is built the way it is.

> *"The publisher, like OpenAI, Meta, Microsoft, xAI, should not be a decision factor."*
> — Microsoft Azure Architecture Center, on model selection criteria (2026)
>
> *"The hybrid approach — combining fine tuning with RAG retrieval — typically outperforms either method in isolation."*
> — Databricks Blog, "RAG vs Fine-Tuning" (2026)

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When Things Don’t Go as Planned

If the diagnostic doesn't cleanly point to RAG, fine-tuning, or prompting, default to prompting and RAG first, since they're cheaper to reverse, and revisit fine-tuning once real usage data shows it's still needed. Likewise, if the team lacks the expertise to evaluate model/provider criteria, that's a signal to lean toward Embedded or BYOAI rather than a custom build, and to get outside review before committing.

What You've Achieved

With the architecture decision record in hand, the organization has moved from an implicit, easily-forgotten technical choice to a documented one — each with its reasoning attached. That record is what makes the system's design legible to the next person who has to maintain, audit, or extend it.

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PART II : AI Adoption Framework

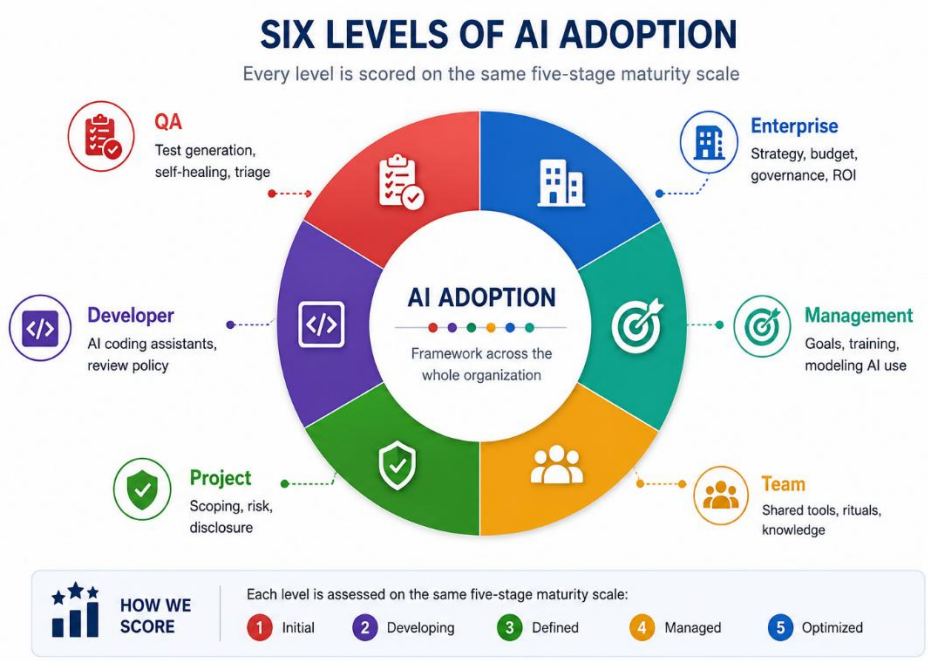
The core maturity model: six organizational levels, a shared five-stage scale, lightweight governance, and stage-by-stage diagnostic questions to find out exactly where the organization stands today. This is the part of the journey Mylar reached once its support pilot worked: a single successful pilot does not make an organization AI-mature, and the same six levels and five stages apply whether the organization has 8 developers or 800.

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Purpose and Scope

This framework gives the organization a shared, repeatable way to introduce, govern, and measure AI adoption across every layer of the business — from boardroom strategy down to the code a developer ships and the tests QA runs.

It is built to serve several goals at once, because in practice they are not separate initiatives but the same initiative viewed from different angles:



1. **Productivity and efficiency:** reduce toil and speed up delivery by putting the right AI tool in the right hands.
2. **Risk and governance:** prevent unmanaged "shadow AI" use, protect data and IP, and keep a human accountable for every AI-assisted decision.
3. **Competitive necessity:** close the gap with faster-moving competitors without over-rotating on hype.
4. **Cost efficiency:** get measurable return from AI spend rather than paying for tools nobody uses well.
5. **Fundamentals:** every level gets a baseline of practical skill and policy, not just the levels closest to the technology.

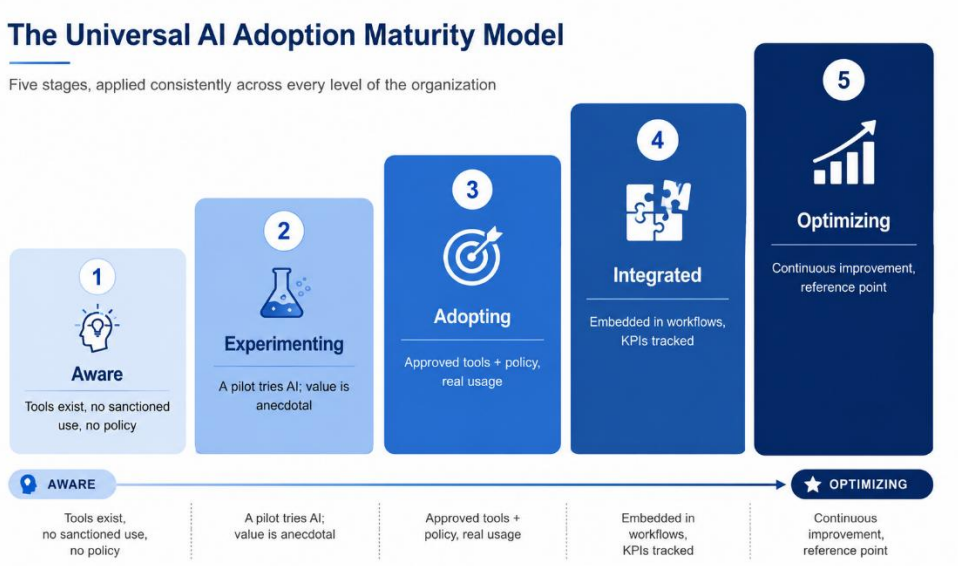
Six organizational levels are in scope: Enterprise, Management, Team, Project, Developer, and QA. Because the organization is small (under 100 people), governance is deliberately lightweight — one person can hold several of these "levels" at once. The structure is designed to scale without a rebuild as the company grows.

New in this version: each level opens with a mind map of what's inside its section, and every one of the five maturity stages now carries five diagnostic questions — each with a concrete example and an explanation of what the answer tells you. A single running illustrative example, "Mylar Technologies" (a fictional small scale SaaS company), is used throughout to show what each stage looks like in practice. Mylar Technologies is a composite for illustration, not a verified real-world case study.

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The Universal AI Adoption Maturity Model

Every level in this framework is measured against the same five-stage maturity scale, so progress is comparable across the whole organization even though the day-to-day work looks different at each level. This mirrors the structure used in Accenture / Carnegie Mellon SEI's 2026 AI Adoption Maturity Model and the AI-native QA maturity models now common in the industry.



Stage	Name	What it looks like
1	Aware	People know AI tools exist and have opinions about them, but there is no sanctioned use, no policy, and no budget. Any use is ad hoc and invisible to leadership.
2	Experimenting	A few individuals or one pilot team try AI tools on real work. No formal policy yet, but leadership is aware and tracking informally. Value is anecdotal.
3	Adopting	A tool set has been chosen and approved. A basic policy exists. Most people at this level use AI regularly on real work, and value starts being tracked with real numbers, not anecdotes.
4	Integrated	AI use is embedded in standard workflows and templates, not a bolt-on. Cross-level consistency exists. KPIs are tracked routinely and reviewed on a cadence.
5	Optimizing	The organization continuously improves how it uses AI: workflows are re-engineered around AI capability, usage data drives decisions, and this level is a source of institutional knowledge for the rest of the company.

How to use this table: every level below is scored on this same 1–5 scale, and the diagnostic questions in each level's section are the practical tool for deciding which stage you're actually in. A realistic target for a funded small organization is Stage 3–4 within 6–12 months of starting deliberate adoption, reaching Stage 5 only in the levels closest to day-to-day technical delivery (Developer, QA) where the tools are most mature.

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The adoption pyramid

Maturity is not evenly distributed across the organization, and it shouldn't be — the levels closest to day-to-day technical delivery move fastest, and the levels that set strategy move more slowly but carry more weight per decision. The pyramid below is an illustrative snapshot of that shape, not a target every organization must reproduce exactly.

Lightweight Governance Foundation (applies enterprise-wide)

Before scoring any level, put a minimal governance foundation in place. For an organization under 100 people this should take days, not months.

Recommended governance pillars

Adapted from 2026 mid-market AI governance guidance — select the ones relevant to your industry rather than implementing all nine at once:

1. Fairness and bias — are you using AI in hiring, lending, or other decisions about people? If yes, this becomes high priority.
2. Reliability and validity — someone checks AI output before it ships, always.
3. Privacy and security — what data is allowed into which AI tool (customer PII, source code, financials).
4. Transparency — you can explain, in plain language, where AI touched a work product.
5. Accountability — a named human owns every AI-assisted decision; AI is never the accountable party.
6. Legal and regulatory compliance — sector-specific rules (e.g., healthcare, finance) reviewed before rollout.
7. Vendor and third-party risk — data handling terms of any AI vendor are reviewed before adoption.
8. Acceptable IP use — policy on AI-generated content ownership and attribution.
9. Tool allow-list — a short, explicit list of approved AI tools; everything else requires sign-off.

Minimal roadmap

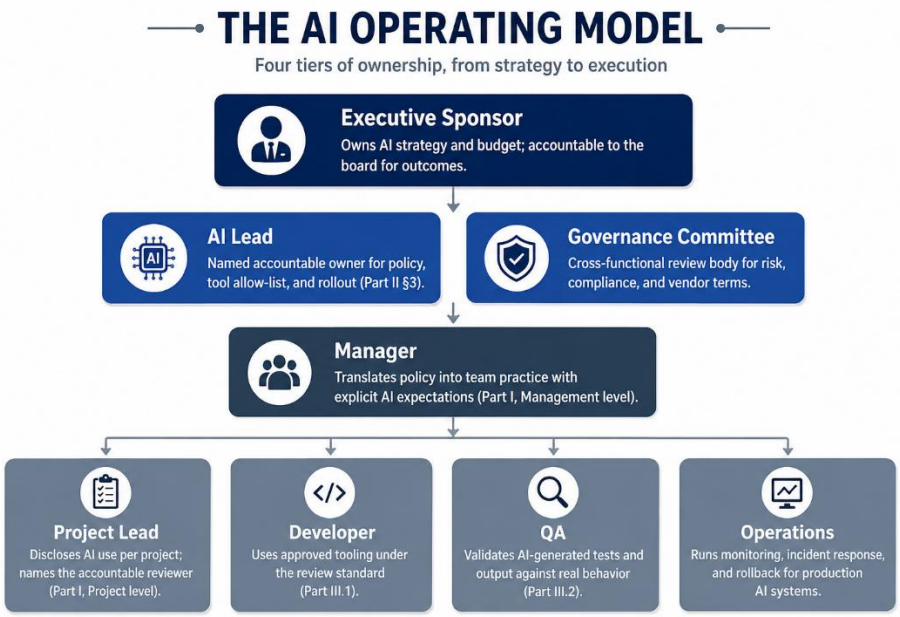
1. Name one person as the AI lead (can be an existing role, e.g., CTO or Head of Ops).
2. Write a one-page “AI dos and don’ts” — what data can go into AI tools, which tools are approved.
3. Publish the approved tool list and require sign-off for anything else.
4. Review the policy quarterly as tools and regulations change.

This foundation is a prerequisite for every level below reaching Stage 3 (“Adopting”) — you cannot have consistent adoption without an approved tool list and a basic policy.

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AI Operating Model & Role Responsibility Matrix

"Who owns AI" was named as one of the five sources of confusion in the Introduction. This section answers it directly with a operating model: a small set of roles, arranged in tiers, each with a specific job in the transformation — and a responsibility matrix that makes explicit who is Responsible, Accountable, Consulted, or Informed for the decisions that come up most often.



1. **Executive Sponsor:** Owns AI strategy and budget; accountable to the board for outcomes.
2. **AI Lead:** Named accountable owner for policy, tool allow-list, and rollout (Chapter 10).
3. **Governance Committee:** Cross-functional review body for risk, compliance, and vendor terms.
4. **Manager:** Translates policy into team practice; sets explicit AI expectations (Chapter 12 (Management Level)).
5. **Project Lead:** Discloses AI use per project; names the accountable reviewer (Chapter 14 (Project Level)).
6. **Developer:** Uses approved tooling under the review standard (Chapter 21).
7. **QA:** Validates AI-generated tests and output against real behavior (Chapter 22).
8. **Operations:** Runs monitoring, incident response, and rollback for production AI systems.

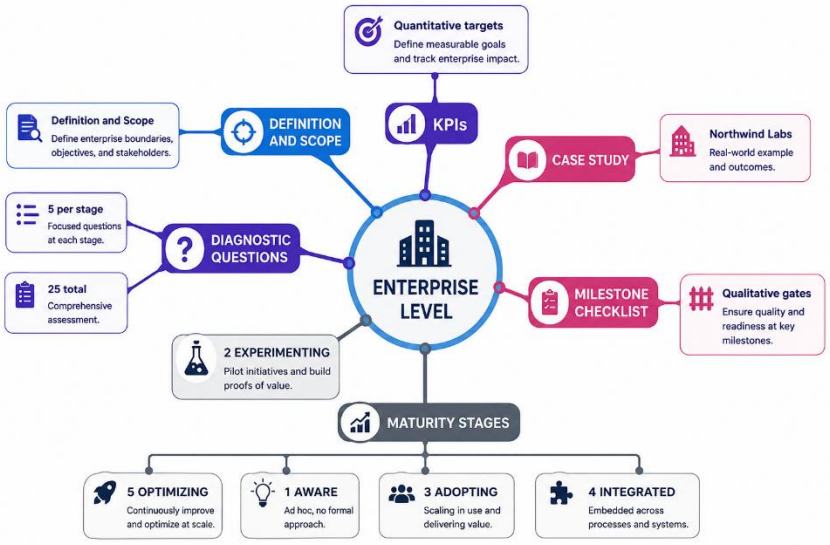
Role Responsibility Matrix (RACI)

R = Responsible (does the work), A = Accountable (owns the outcome — exactly one per row), C = Consulted (input sought beforehand), I = Informed (told after the fact), "–" = not involved in this decision.

Level-by-Level Breakdown

Each level below opens with a mind map of what's inside its section, then covers: what the level covers, the five maturity stages with five diagnostic questions each (with an example and explanation per question), a milestone checklist, the quantitative KPIs used to confirm progress, and a Mylar Technologies case-study snapshot at each stage.

Chapter 11: Enterprise Level



Covers overall AI strategy, budget, tool/vendor selection, cross-functional governance, culture, and ROI reporting to leadership or the board.

Setting the Scene

Enterprise-level AI adoption sets the tone for everything below it: the budget it does or doesn't get, the risk appetite leadership is willing to state out loud, and whether every other level in this book has air cover to move. Get this level wrong and no amount of team-level discipline compensates for it.

Where Most Teams Start

Most enterprises start with AI as a side conversation — a board mention, an executive's personal experiment, a competitor's press release — with no named owner, no budget line, and no shared definition of what "AI-ready" would even mean.

A Better Approach

The better approach treats enterprise AI adoption as a maturity journey to manage deliberately, on the same five-stage ladder used at every other level, so leadership can point to exactly where the organization stands rather than to a feeling.

Foundational Concepts

The Enterprise level is where AI adoption starts as a deliberate choice rather than something that happens to the organization by accident. It covers the strategy, budget, tool selection, and governance decisions that everything else in this framework depends on.

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What This Section Helps You Achieve

The goal is a company that treats AI the way it treats any other strategic capability: funded, owned by a named person, measured, and reviewed on a regular cadence — not an accident of individual enthusiasm.

The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

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Step by Step — The Five Maturity Stages

The rest of this section works through each maturity stage in depth: a short description, then five diagnostic questions with an example and explanation for each, to help you pin down exactly which stage you're in. Each stage opens with the same five-stage roadmap, with the current stage highlighted, so it's always clear where a given stage sits relative to the whole scale.

Stage 1 — Aware



No AI strategy; leadership discusses AI informally.

Key questions to consider

1. **Has leadership ever formally discussed AI as a business topic (not just informally at lunch)?**
   Example: *The founders mention AI in a hallway conversation, but it's never on a board agenda.* **Explanation:** If AI has never reached a formal agenda, the enterprise is at Aware — awareness without action.
2. **Does anyone in the company know what data leaves the building when someone pastes into an AI chatbot?**
   Example: *An employee pastes a customer contract into a public AI chatbot to summarize it.* **Explanation:** No one tracking this signals zero governance — a hallmark of Stage 1.
3. **Is there a budget line item, however small, for AI tools?**
   Example: *No line item exists; any AI subscription is an individual expensing a $20/month tool.* **Explanation:** Absence of any earmarked spend confirms there's no sanctioned enterprise commitment yet.
4. **Could you name one specific business problem AI might solve for this company?**
   Example: *Leadership says "AI is important" but can't name a concrete use case.* **Explanation:** Vague enthusiasm without a specific problem statement is classic Stage 1 thinking.
5. **Has anyone been assigned even informal responsibility for AI questions?**
   Example: *If someone asks "can I use this AI tool," there's no clear person to ask.* **Explanation:** No owner, even an unofficial one, means accountability hasn't started forming.

Stage 2 — Experimenting



One pilot use case funded; no enterprise budget line.

Key questions to consider

1. **Is there at least one funded pilot using AI on real business work?**

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Example: *The ops team trials an AI tool to draft customer support replies for two weeks.* **Explanation:** A single funded pilot is the entry ticket to Stage 2 — intent has become action.

2. **Does leadership know the pilot is happening, even if they aren't running it?**
   Example: *The CEO knows the support team is testing an AI tool and asked for a monthly update.* **Explanation:** Awareness without formal oversight is exactly the boundary between Stage 1 and 2.
3. **Is anyone informally tracking whether the pilot is helping?**
   Example: *The pilot lead keeps a rough note of "felt faster" without real numbers.* **Explanation:** Value being anecdotal, not measured, is the signature of Experimenting rather than Adopting.
4. **Has the pilot surfaced any risk or data concern yet?**
   Example: *The team realizes the AI tool stores prompts on a third-party server.* **Explanation:** Surfacing, but not yet resolving, risk is typical groundwork happening at this stage.
5. **If the pilot ended tomorrow, would the company miss it?**
   Example: *The team says "it was nice to have" but could easily revert to the old way.* **Explanation:** Low dependency confirms the tool hasn't become load-bearing yet — still exploratory.

Stage 3 — Adopting



Written AI strategy and budget exist; approved tool list published; AI lead named.

Key questions to consider

1. **Is there a written, approved list of AI tools employees are allowed to use?**
   Example: *An internal doc names three approved tools and states everything else needs sign-off.* **Explanation:** A published allow-list is the clearest marker that Stage 3 has been reached.
2. **Does the company have a named AI lead — someone accountable for policy and rollout?**
   Example: *The CTO is designated as the AI lead and reviews new tool requests.* **Explanation:** Naming an accountable owner turns ad hoc governance into a real function.
3. **Is there a real budget line for AI tools and training?**
   Example: *The next fiscal year's budget includes a specific "AI tools & training" line.* **Explanation:** Dedicated budget signals the company is committing resources, not just tolerating use.
4. **Are more than half the relevant employees actually using an approved tool regularly?**
   Example: *Six of ten customer-facing staff use the approved AI drafting tool weekly.* **Explanation:** Majority real usage — not just policy on paper — is what separates Stage 3 from Stage 2.
5. **Is anyone tracking a real number (cost, time, or output) tied to AI use?**
   Example: *The team tracks average response-drafting time before and after adoption.* **Explanation:** The first real metric, even a rough one, marks the shift from anecdote to evidence.

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Stage 4 — Integrated



AI strategy tied to business goals; quarterly ROI reporting; policy enforced across all levels below.

Key questions to consider

1. **Is AI use written into standard operating procedures or templates, not just a policy memo?**
   Example: *The onboarding template and process docs reference which AI tools to use and when.* **Explanation:** Embedding AI into the actual templates people use daily is what "integrated" means.
2. **Is AI ROI reported on a regular cadence to leadership, not just when someone asks?**
   Example: *A quarterly leadership deck includes a slide on AI cost savings and adoption rates.* **Explanation:** Routine reporting demonstrates the metric has become part of how the business runs.
3. **Does the AI policy apply consistently across departments, not just the pilot team?**
   Example: *The same tool allow-list and review policy governs support, sales, and engineering.* **Explanation:** Cross-functional consistency is required to call the enterprise level integrated.
4. **Are hiring or vendor decisions ever influenced by AI capability considerations?**
   Example: *A new vendor contract is evaluated partly on whether it supports the approved AI stack.* **Explanation:** AI factoring into unrelated decisions shows it has become structural, not incidental.
5. **Could a new employee learn the AI policy from documentation alone, without asking around?**
   Example: *A new hire reads the onboarding wiki and understands exactly what's approved.* **Explanation:** Self-service clarity is a proxy for how deeply the policy has been operationalized.

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Stage 5 — Optimizing



AI strategy actively reshapes how the business operates; enterprise is a reference point others benchmark against.

Key questions to consider

1. **Has AI usage data ever changed a strategic decision (a product, a headcount plan, a market)?**
   Example: *Adoption data showed AI cut support-ticket handling time so much that leadership redirected two planned support hires into a new customer-success role instead.* **Explanation:** Data-driven strategic pivots are the clearest sign of Stage 5 maturity.
2. **Does the company get asked by peers or partners how it approaches AI?**
   Example: *A partner company asks Mylar Technologies to share its AI governance policy as a template.*
   Explanation: Being sought out as a reference point is an external validation of leading practice.
3. **Is there a continuous improvement loop — policy or tool choices revised based on evidence?**
   Example: *The approved tool list changes quarterly based on measured performance, not vendor pitches.*
   Explanation: A living, evidence-driven policy, not a static one, is what "continuously improving" looks like.
4. **Has the company redesigned a workflow specifically because AI made a new approach possible?**
   Example: *Support tickets are now triaged and routed automatically — a workflow that didn't exist pre-AI.*
   Explanation: True re-engineering, not just speeding up the old workflow, marks the top stage.
5. **If AI tools disappeared tomorrow, would it seriously disrupt how the business runs?**
   Example: *Losing the AI stack would force the company to rehire for roles it has redesigned around AI.*
   Explanation: High dependency, matched with strong governance, indicates deep, healthy integration rather than fragile hype-chasing.

What It Looks Like in Practice

Milestone checklist

1. AI lead named
2. Written strategy exists
3. Approved tool list published
4. Budget line exists
5. Quarterly review cadence set

KPIs (quantitative)

1. % of AI spend mapped to a measured business outcome
2. Number of approved tools in active use
3. Cost saved or revenue enabled per quarter attributable to AI

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4. Enterprise maturity stage (1–5), tracked quarterly

When Things Don’t Go as Planned

If no executive will sponsor AI formally yet, don't wait for a mandate that may never come — start the readiness assessment and the governance foundation informally, using Part I's tools, and let an early, well-measured win earn the sponsorship. Revisit formal executive ownership once that proof point exists.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study below, this level's journey runs from a hallway conversation between two co-founders to a named AI lead and a standing quarterly ROI slide within about a year.

> *“The SEI AI adoption maturity assessment provided far more than a point-in-time evaluation—it gave us a structured, actionable understanding of where we are succeeding, where more attention may be needed, and how to prioritize future investments for maximum ROI.”*
> — Srinivasulu Nasam, Head of Enterprise AI Transformation, BGSW — quoted in the SEI/Accenture AI Adoption Maturity Model announcement (2026)
>
> *“Successful AI adoption goes beyond improving automation or augmenting existing processes. It means rethinking workflows and innovating ways to bolster them with AI.”*
> — Ipek Ozkaya, Technical Director, SEI AI-Native Software Engineering — same announcement

Case study: Mylar Technologies (illustrative composite)

Mylar Technologies is a fictional small scale software company building custom scheduling and workflow integrations for mid-market clients, used throughout this framework as a running, illustrative example — the same company whose support-ticket pilot was introduced in the Executive Summary and worked through Part I. It is a composite, not a verified real-world case, built to show how one organization's practice might look at each stage.

Stage	Mylar Technologies at this stage
1 Aware	AI came up only in hallway chats between the two co-founders; no formal discussion, no budget, no policy.
2 Experimenting	Support Ops quietly piloted an AI drafting tool for two weeks; the CEO knew about it and asked for an update, but nothing was tracked formally.
3 Adopting	Mylar published its first approved-tools list, named the VP of Engineering as AI lead, and added a small "AI tools & training" line to next year's budget.
4 Integrated	AI ROI became a standing slide in the quarterly all-hands deck, and the same tool policy now applies to Support, Sales, and Engineering alike.
5 Optimizing	Adoption data showed AI cut support-ticket handling time so much that leadership redirected two planned support hires into a new customer-success role instead — and a partner company later asked to copy Mylar's AI governance policy.

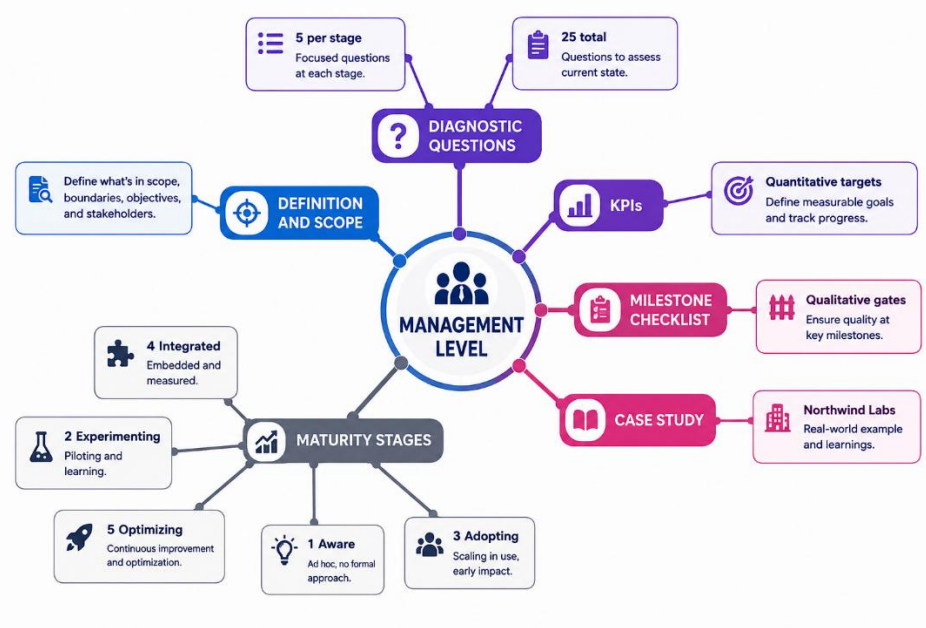
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What You've Achieved

You'll know the Enterprise level has matured when AI decisions stop needing a special meeting — they happen inside the normal budget cycle, hiring process, and quarterly review, the same way any other core business capability would.

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Chapter 12: Management Level



Covers how managers set goals, allocate resources, train their people, and model AI use themselves.

Setting the Scene

Managers are where enterprise intent either becomes daily practice or quietly dies. A strategy document means nothing until managers are trained, resourced, and willing to actually apply the policy to their own team's work.

Where Most Teams Start

Most managers first encounter AI policy the same way they encounter most compliance memos: an email to skim, no training, and no real sense of what's expected of them day to day — so it gets filed and forgotten until an incident forces the question.

A Better Approach

The better approach makes management-level adoption a trained, tracked practice — managers who can explain the policy, apply it to real decisions, and are measured on doing so — rather than a memo assumed to be self-executing.

Foundational Concepts

The Management level is where enterprise policy either becomes real daily behavior or stays a document nobody reads. Managers are the translation layer between a company-wide AI policy and what actually happens on a team.

What This Section Helps You Achieve

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The goal is for every manager to give their people clear, consistent guidance on AI use and to model good practice themselves, so adoption doesn't depend on which manager a given employee happens to have.

The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

Step by Step — The Five Maturity Stages

The rest of this section works through each maturity stage in depth: a short description, then five diagnostic questions with an example and explanation for each, to help you pin down exactly which stage you're in. Each stage opens with the same five-stage roadmap, with the current stage highlighted, so it's always clear where a given stage sits relative to the whole scale.

Stage 1 — Aware



Managers have not discussed AI use with their teams.

Key questions to consider

1. **Have you (as a manager) ever brought up AI tools in a 1:1 or team meeting?**
   Example: *A manager has never mentioned AI to their reports.* **Explanation:** Silence on the topic is the default Stage 1 state.
2. **Do you know which AI tools, if any, your direct reports already use unofficially?**
   Example: *A report has been using a personal chatbot account for months; the manager finds out by accident.* **Explanation:** Not knowing about existing shadow use means there's no visibility yet.
3. **Is AI mentioned anywhere in your team's goals or OKRs?**
   Example: *This quarter's goals make no reference to AI tools or efficiency gains from them.* **Explanation:** Absence from goal-setting confirms it isn't yet a management lever.
4. **Would you know who to ask if a report asked, "can I use AI for this task?"**
   Example: *The manager would have to guess or escalate randomly.* **Explanation:** No clear path to an answer reflects the lack of structure typical of Stage 1.
5. **Have you personally tried an AI tool for your own management work?**
   Example: *The manager has never tried using AI to draft a performance review.* **Explanation:** Personal non-use often mirrors and reinforces team-level non-use.

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Stage 2 — Experimenting



A manager encourages one or two direct reports to try a tool informally.

Key questions to consider

1. **Have you encouraged at least one report to try an AI tool on a real task?**
   Example: *A manager suggests a report try an AI tool to draft a project update.* **Explanation:** A single nudge is the entry point to Stage 2.
2. **Have you tried using AI yourself for a management task and noticed a difference?**
   Example: *The manager drafts a review with AI assistance and finds it saves 20 minutes.* **Explanation:** Personal firsthand experience typically precedes team-wide encouragement.
3. **Has AI come up organically in a team meeting, even briefly?**
   Example: *Someone mentions in standup that AI helped them finish a task early.* **Explanation:** Organic mentions show awareness is spreading beyond leadership.
4. **Do you have even an informal sense of who on your team is using AI and who isn't?**
   Example: *The manager can name two of six reports using AI tools regularly.* **Explanation:** Informal tracking, even incomplete, is the hallmark of this stage.
5. **Has a report asked you a question about AI use that you didn't have a ready answer for?**
   Example: *A report asks if it's okay to paste client data into an AI tool, and the manager has to say "let me find out."* **Explanation:** Surfacing unanswered questions is normal groundwork at Stage 2.

Stage 3 — Adopting



Managers set explicit expectations ("use AI for X, not for Y"), and AI use appears in 1:1s.

Key questions to consider

1. **Have you set an explicit expectation about when AI should (and shouldn't) be used on your team?**
   Example: *The manager tells the team, "use AI for first drafts, but final client communication needs a human review."* **Explanation:** An explicit rule is the marker of Stage 3.
2. **Does AI use come up as a normal topic in 1:1s, not just when problems arise?**
   Example: *"How's the AI tool working for you" becomes a normal 1:1 check-in question.* **Explanation:** Routine, positively framed check-ins show it has become a managed topic.

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3. **Would most of your reports say they know what's expected of them regarding AI use?**
   Example: *In a quick poll, five of six reports can describe the team's AI expectations accurately.* **Explanation:** Majority clarity confirms adoption, not just a policy that exists on paper.
4. **Have you flagged any AI-related risk or misuse to the AI lead / enterprise policy owner?**
   Example: *The manager notices a report using an unapproved tool and reports it to the AI lead.* **Explanation:** Two-way communication with enterprise governance shows the management layer is functioning as intended.
5. **Do you factor AI fluency into how you evaluate performance, even informally?**
   Example: *A performance conversation includes a note that a report is a fast, thoughtful adopter of the new tools.* **Explanation:** Weaving it into evaluation, even informally, marks real behavioral adoption.

Stage 4 — Integrated



AI fluency is part of onboarding, performance conversations, and hiring criteria for the role.

Key questions to consider

1. **Is AI fluency an explicit part of your onboarding checklist for new hires?**
   Example: *New hires get a specific onboarding module on approved AI tools and expectations.* **Explanation:** Formal onboarding inclusion is the clearest Stage 4 signal.
2. **Is AI fluency a stated criterion when you hire for this role?**
   Example: *A job posting for a new team member lists "comfort with AI-assisted workflows" as a requirement.* **Explanation:** Hiring criteria reflect that AI fluency is now a baseline expectation, not a bonus.
3. **Do performance review templates include a section touching on AI-assisted work quality?**
   Example: *The review template asks managers to comment on how well someone integrates AI into their workflow.* **Explanation:** Institutionalizing it in the template, not just verbally, shows true integration.
4. **Could another manager cover your team for a week and enforce the same AI expectations without you briefing them?**
   Example: *A peer manager fills in and naturally continues the same AI practices because they're documented.* **Explanation:** Portability of practice indicates it's integrated into the team's operating system, not personality-dependent.
5. **Do you regularly compare team output metrics from before and after AI adoption?**
   Example: *The manager reviews a quarterly chart of ticket resolution time pre/post AI rollout.* **Explanation:** Routine, quantified comparison is a hallmark of Stage 4 discipline.

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Stage 5 — Optimizing



Managers actively redesign team workflows around AI capability and coach others on doing the same.

Key questions to consider

1. **Have you redesigned a workflow (not just sped up an old one) because of what AI makes possible?**
   Example: *Instead of writing first-draft replies faster, the team now has AI triage and route tickets automatically, changing the reports' job itself.* **Explanation:** Redesigning the work, not just accelerating it, is the top-tier signal.
2. **Do other managers ask you how you run your team's AI practice?**
   Example: *A peer manager copies this manager's prompt library and ritual for their own team.* **Explanation:** Being emulated by peers indicates leading-edge practice.
3. **Do you coach your reports on AI practice the way you'd coach any other core skill?**
   Example: *AI prompt technique becomes a regular coaching topic, like coaching someone's writing or review skills.*
   Explanation: Treating it as a first-class coachable skill shows deep, sustained investment.
4. **Has your team's AI practice measurably changed a business outcome leadership cares about?**
   Example: *The manager can point to a specific metric (retention, revenue, cost) that moved because of how the team uses AI.* **Explanation:** Tying practice to bottom-line outcomes is the clearest proof of Stage 5.
5. **If a new AI tool appeared tomorrow, would your team have a habit-based process to evaluate and adopt it quickly?**
   Example: *The team has a lightweight "try, measure, decide" ritual used whenever a new tool shows up.*
   Explanation: A repeatable adoption muscle, not one-off enthusiasm, defines an optimizing team.

What It Looks Like in Practice

Milestone checklist

1. Managers trained on the AI policy
2. AI use discussed in team meetings
3. AI fluency added to onboarding
4. A manager is the point of escalation for AI questions

KPIs (quantitative)

1. % of managers who have completed AI policy training
2. % of teams with a documented AI use expectation
3. Employee survey score on "my manager gives clear guidance on AI use"

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When Things Don’t Go as Planned

If formal manager training isn't budgeted yet, don't let the policy sit unenforced in the meantime — have one manager pilot it, document what worked, and use that as the training material once budget opens up. A single well-documented example beats waiting for a company-wide rollout.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study, the Head of Support moves from not knowing her reps used AI at all, to an explicit team rule, to being copied by peer managers across the company.

> *“Within your governance team, think about your AI dos and don'ts. What do you want to use AI for — and what do you not want it used for?”*
> — Theron Kelso, Wipfli — AI Governance Checklist for Mid-Market Companies (2026)
>
> *“This typically includes your CFO, CIO and COO, although larger organizations might delegate some of the responsibilities to an AI governance body or working group.”*
> — Theron Kelso, Wipfli — on who ends up owning this guidance in a smaller organization

Case study: Mylar Technologies (illustrative composite)

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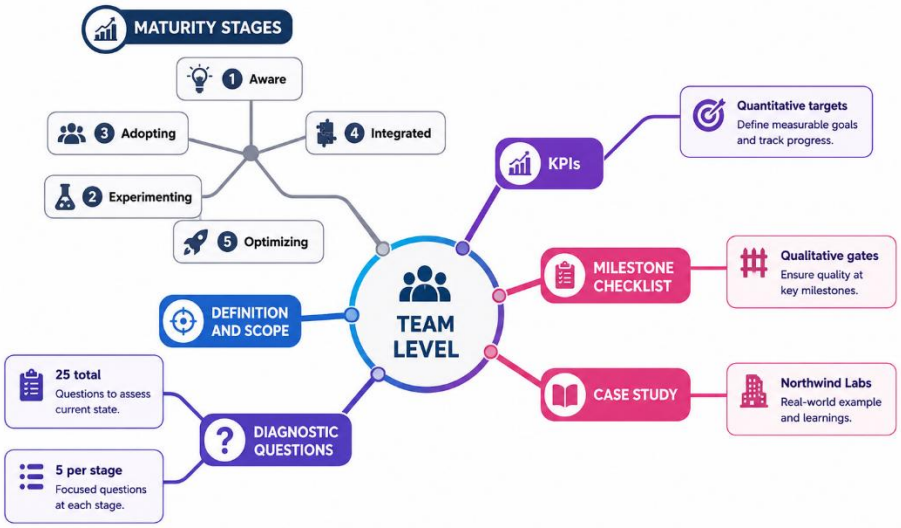
Stage	Mylar Technologies at this stage
1 Aware	The Head of Support has never mentioned AI tools to her team, and doesn't know two reps already use a free chatbot on the side.
2 Experimenting	She notices the two reps using AI and, intrigued, encourages a third to try it on ticket drafts.
3 Adopting	She sets an explicit rule — AI for first drafts, human review for anything client-facing — and starts asking about it in 1:1s.
4 Integrated	AI fluency becomes part of the onboarding checklist for new support hires, and review templates now ask managers to comment on AI-assisted work quality.
5 Optimizing	Other team leads at Mylar copy her prompt library, and she can point to a specific metric — average first-response time — that improved because of how her team uses AI.

What You've Achieved

You'll know management-level adoption has matured when a new manager can step into any team and find the same clear AI expectations already documented and modeled — nothing depends on one particular person's personal initiative.

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Chapter 13: Team Level



Covers shared practices across a functional or cross-functional team (e.g., a product squad) — common tools, knowledge sharing, rituals.

Setting the Scene

Teams are where AI adoption becomes visible in actual output — shared workflows, common tools, and whether one team's practice spreads to another or stays trapped with a single enthusiast.

Where Most Teams Start

Most teams start with one person quietly using an AI tool on their own initiative, with teammates only vaguely aware of it and no shared workflow, so the practice never outlives that one person's attention or tenure.

A Better Approach

The better approach makes AI practice a team-level asset — shared tools, a common workflow, and knowledge that survives any one person leaving — rather than a skill that lives and dies with an individual.

Foundational Concepts

The Team level is where AI adoption becomes a shared habit instead of an individual quirk. It's the difference between one person quietly using a tool and a team that has agreed on how it works together.

What This Section Helps You Achieve

The goal is a team with a common toolset, a place to share what's working, and at least one standing ritual that keeps AI practice visible and improving, rather than depending on one enthusiastic teammate.

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The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

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Stage 1 — Aware



No shared team practice; everyone does their own thing.

Key questions to consider

1. **Does everyone on the team use a different (or no) approach to AI, with no shared practice?**
   Example: *One person uses one assistant, another uses nothing, a third uses a random free tool.* **Explanation:** Total inconsistency with no shared reference point is classic Stage 1.
2. **Has AI ever come up in a team retro or planning meeting?**
   Example: *Six months of retros pass with zero mention of AI.* **Explanation:** Total absence from team rituals confirms no shared awareness yet.
3. **Is there a shared place (channel, doc) where the team could even put AI tips if someone had one?**
   Example: *No such channel or doc exists.* **Explanation:** No shared infrastructure means the team has no mechanism for spreading practice.
4. **Would a new team member have any idea AI tools were relevant to this team's work?**
   Example: *Nothing in onboarding or team docs mentions AI.* **Explanation:** Invisibility to newcomers confirms it isn't part of team identity yet.
5. **Has the team ever discussed whether AI could help with a recurring pain point?**
   Example: *The team complains weekly about a repetitive task but never connects it to AI.* **Explanation:** Not connecting a known pain point to a known solution is a hallmark of Stage 1.

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Stage 2 — Experimenting



A couple of team members share tips informally.

Key questions to consider

1. **Have at least two team members swapped a tip about an AI tool informally?**
   Example: *Two teammates chat in Slack DMs about a useful prompt.* **Explanation:** Peer-to-peer informal sharing is the seed of Stage 2.
2. **Has anyone brought up an AI win in a stand-up or retro, even casually?**
   Example: *Someone mentions offhand, "AI helped me finish that doc faster."* **Explanation:** Surfacing a win in a team setting, not just DMs, marks progress from Stage 1.
3. **Is there rough, shared awareness of who on the team is trying AI tools?**
   Example: *The team lead can name a couple of early adopters.* **Explanation:** Emerging visibility, even informal, is typical at this stage.
4. **Has the team tried using AI on one shared task together, even once?**
   Example: *The team jointly tests an AI tool to draft a shared document.* **Explanation:** A single shared experiment, not just individual use, is the marker of team-level experimenting.
5. **Would the team notice if the one or two AI-experimenting members left?**
   Example: *If the early adopter left, the practice would likely disappear with them.* **Explanation:** Practice tied to one or two people, not the team, shows it's still fragile and personal.

Stage 3 — Adopting



The team has agreed on a common tool set and a place to share prompts/wins.

Key questions to consider

1. **Has the team agreed on a common set of AI tools to use?**
   Example: *The team picks one approved coding or writing assistant as the team standard.* **Explanation:** A shared, agreed tool set is the clearest Stage 3 signal.
2. **Is there a live place (channel or wiki page) where prompts and wins get shared regularly?**
   Example: *A dedicated tips channel gets a new post every week or two.* **Explanation:** An active, ongoing sharing habit, not a one-off, marks real adoption.

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3. **Do most team members use an approved tool weekly?**
   Example: *Seven of nine team members log into the approved tool at least weekly.* **Explanation:** Majority regular use is the bar for calling a team "adopting."
4. **Has the team measured anything about its own AI use, even roughly?**
   Example: *The team informally notes that draft turnaround time seems faster.* **Explanation:** The first, even rough, self-measurement shows evidence is starting to matter.
5. **If a new member joined, would the existing team walk them through the shared AI practice?**
   Example: *Onboarding a new teammate includes a ten-minute intro to the team's AI tools and channel.*
   Explanation: The practice becoming teachable to newcomers shows it has become a real team norm.

Stage 4 — Integrated



AI use is part of team rituals — a recurring "AI show and tell," shared prompt library, retro discussion.

Key questions to consider

1. **Is there a recurring ritual (show-and-tell, retro item) specifically about AI use?**
   Example: *Every other retro includes a five-minute "AI win of the sprint" segment.* **Explanation:** A standing ritual embeds AI into the team's regular rhythm, not just ad hoc chat.
2. **Does the team have an actual shared prompt library or playbook, not just a chat thread?**
   Example: *A maintained doc lists the team's ten best prompts for common tasks.* **Explanation:** A structured, discoverable artifact, versus scattered chat messages, is what "integrated" looks like.
3. **Can the team point to a real before/after number tied to its AI practice?**
   Example: *The team tracks that average review turnaround dropped by a day after adopting an AI review assistant.* **Explanation:** A tracked, cited metric confirms the practice has moved past anecdote.
4. **Would a manager reviewing the team's workflow see AI baked into the standard process, not bolted on?**
   Example: *The team's documented workflow diagram includes an AI-assisted step as a normal stage, not a footnote.* **Explanation:** Appearing in the actual process documentation is the test for integration.
5. **Has another team asked to borrow this team's AI practice?**
   Example: *A sibling team asks to copy this team's prompt library.* **Explanation:** External interest from peers is a strong proxy that the practice is mature and portable.

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Stage 5 — Optimizing



The team's AI practices are documented and adopted by other teams; measurably faster or fewer defects.

Key questions to consider

1. **Has the team's AI practice been written up and adopted org-wide?**
   Example: *The team's prompt library becomes the company standard used by three other teams.* **Explanation:** Org-wide adoption of a team-originated practice is the clearest Stage 5 marker.
2. **Does the team measurably outperform comparable teams on speed or quality, attributable partly to AI practice?**
   Example: *This team's cycle time is 30% faster than a similar team, partly attributed to its mature AI workflow.*
   Explanation: A measurable, attributable performance gap is strong evidence of optimization.
3. **Does the team proactively retire or replace AI tools/practices based on evidence, not habit?**
   Example: *The team drops a tool it had used for a year after data showed a newer one performed better.*
   Explanation: Willingness to change based on evidence, not attachment, shows a mature, optimizing mindset.
4. **Has the team redesigned how it works, not just sped up old steps, because of AI?**
   Example: *Review became an AI-assisted triage step that changes what human reviewers actually spend time on.*
   Explanation: Structural redesign, not just faster old steps, is the top-tier signal.
5. **Could this team run a workshop teaching its AI practice to another team?**
   Example: *The team runs a lunch-and-learn that other teams attend to learn its practices.* **Explanation:** Being capable of formally teaching the practice shows the team has turned tacit skill into transferable knowledge.

What It Looks Like in Practice

Milestone checklist

1. Shared tool list agreed
2. A knowledge-sharing channel exists
3. At least one recurring ritual (show-and-tell, retro item)
4. Team-level KPIs tracked

KPIs (quantitative)

1. Cycle time (lead time from start to done) before vs. after adoption
2. Number of shared prompts/playbooks in the team's library
3. % of team members actively using an approved tool weekly

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When Things Don’t Go as Planned

If the team can't agree on a shared tool yet, don't force premature standardization — let the pilot users document what's actually working, and use that evidence (not seniority or preference) to make the shared choice once there's something real to compare.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study, the five-person support pod goes from total inconsistency to becoming the company-wide standard other teams copy.

> *“Successful AI adoption goes beyond improving automation or augmenting existing processes. It means rethinking workflows and innovating ways to bolster them with AI.”*
> — Ipek Ozkaya, SEI — the same idea applied at team scale: a team at Stage 4-5 isn't just working faster, it has reworked how it works

Case study: Mylar Technologies (illustrative composite)

Mylar Technologies is a fictional small scale software company building custom scheduling and workflow integrations for mid-market clients, used throughout this framework as a running, illustrative example. It is a composite, not a verified real-world case, built to show how one organization's practice might look at each stage.

Stage	Mylar Technologies at this stage
1 Aware	The five-person support pod all use different (or no) tools, and AI has never come up in a retro.
2 Experimenting	Two teammates swap a tip about a useful prompt in a Slack DM, and someone mentions an AI win in standup.
3 Adopting	The pod agrees on one shared drafting tool and starts a tips channel that gets a new post most weeks.
4 Integrated	Every other retro now includes a five-minute "AI win of the sprint," and the team's prompt library gets borrowed by a sibling team.
5 Optimizing	The pod's practice becomes the company standard, adopted by two other teams, and the pod's ticket cycle time is measurably faster than a comparable team's.

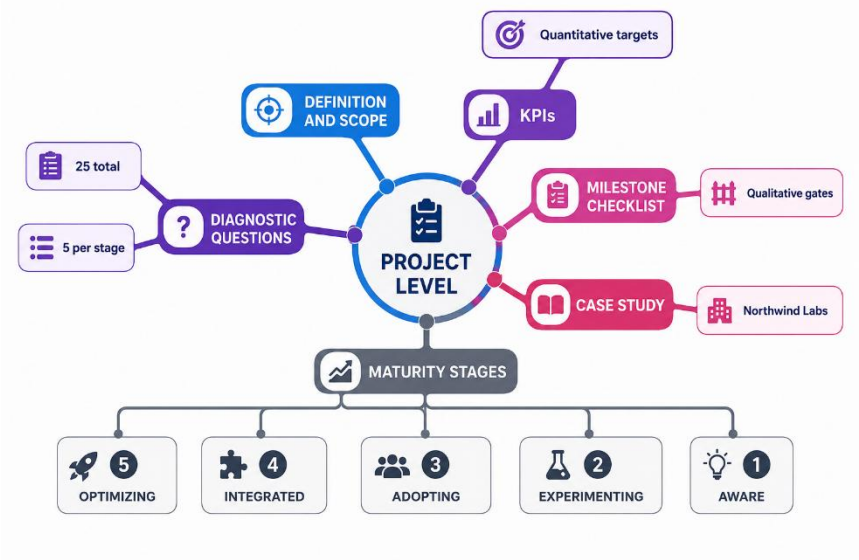
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What You've Achieved

You'll know a team has matured when its AI practice would survive any single person leaving — it's written down, ritualized, and other teams are asking to borrow it.

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Chapter 14: Project Level



Covers how AI use is planned for and accounted for within an individual project's scoping, risk assessment, and documentation.

Setting the Scene

Projects are where AI's use has to be disclosed, reviewed, and stand behind a delivered outcome — this is the level where adoption meets actual client or stakeholder accountability.

Where Most Teams Start

Most projects use AI quietly, without disclosure, and without anyone tracking which parts of the deliverable were AI-assisted — which is fine until a client, auditor, or incident asks the question directly and no one has an answer.

A Better Approach

The better approach builds AI disclosure and review into the project template itself — so every deliverable can state, on request, what AI touched and who reviewed it — rather than reconstructing that answer after the fact under pressure.

Foundational Concepts

The Project level is where AI use gets planned for, not just allowed. It's about whether a project's scope, risk assessment, and sign-off process actually account for AI, the same way they'd account for any other dependency.

What This Section Helps You Achieve

The goal is for AI use to be a visible, disclosed, reviewed part of how a project is planned and closed out — never a silent, undocumented ingredient in the deliverable.

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The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

Step by Step — The Five Maturity Stages

The rest of this section works through each maturity stage in depth: a short description, then five diagnostic questions with an example and explanation for each, to help you pin down exactly which stage you're in. Each stage opens with the same five-stage roadmap, with the current stage highlighted, so it's always clear where a given stage sits relative to the whole scale.

Stage 1 — Aware



Projects are scoped with no mention of AI tooling or risk.

Key questions to consider

1. **Does a typical project plan mention AI anywhere?**
   Example: *The latest project charter has zero references to AI tools or risk.* **Explanation:** Total absence from planning documents is classic Stage 1.
2. **If an AI tool were used on a project, would anyone know to flag it as a risk item?**
   Example: *No one on the project team considers AI use a risk category worth naming.* **Explanation:** Not even recognizing it as a risk category shows no awareness yet.
3. **Has a client or stakeholder ever asked whether AI was used on a deliverable, catching the team off guard?**
   Example: *A client asks, "did you use AI to write this," and the team is caught off guard.* **Explanation:** Being unprepared for an obvious stakeholder question reflects Stage 1.
4. **Does the project retrospective template have any place to discuss AI's role?**
   Example: *The standard retro template has no AI-related question.* **Explanation:** Its absence from the standard template confirms it isn't yet part of the project process.
5. **Could someone name a project where AI could plausibly have helped, even if it wasn't used?**
   Example: *No one on the team has considered where AI might fit into the current project.* **Explanation:** Not even hypothetically considering it is a sign the project level hasn't engaged with AI yet.

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Stage 2 — Experimenting



One project tries an AI-assisted approach informally.

Key questions to consider

1. **Has one project informally tried an AI-assisted approach for part of its work?**
   Example: *One project uses an AI tool to draft a first version of a requirements doc.* **Explanation:** A single informal trial marks the start of Stage 2.
2. **Did that project note anything about the experience afterward, even informally?**
   Example: *The project lead mentions in a retro that AI "saved some time" on the doc.* **Explanation:** An informal after-the-fact note, even without real data, shows early tracking.
3. **Is there any awareness across projects that this experiment happened?**
   Example: *Other project leads have heard about it secondhand.* **Explanation:** Cross-project awareness, even secondhand, shows information is starting to spread.
4. **Did the experiment surface any concern that hasn't been resolved yet?**
   Example: *The team wonders whether AI-drafted content could have IP implications, but no one has answered it.* **Explanation:** Surfacing unresolved questions is typical, expected groundwork at this stage.
5. **If asked, could the project lead describe roughly what worked and what didn't?**
   Example: *The lead can informally say, "good for a first draft, not for final client copy."* **Explanation:** A rough, qualitative read, not hard numbers, fits the Experimenting profile.

Stage 3 — Adopting



Project plans include an explicit line for "where AI is used" and note the accountable reviewer.

Key questions to consider

1. **Does the project plan template include an explicit line for "where AI is used"?**
   Example: *The updated project template has a required field: "AI tools used in this project."* **Explanation:** A required field in the standard template is the clearest Stage 3 marker.
2. **Is there a named accountable reviewer for AI-assisted deliverables on the project?**
   Example: *The project plan names the tech lead as the reviewer for any AI-assisted client deliverable.* **Explanation:** Naming an accountable human is central to responsible adoption at this stage.

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3. **Are most active projects actually filling out that AI disclosure field, not leaving it blank?**
   Example: *Four of five active projects have a completed AI disclosure section.* **Explanation:** Real usage of the new field, not just its existence, shows genuine adoption.
4. **Has at least one project retrospective specifically discussed AI's impact on schedule or quality?**
   Example: *A retro notes, "the AI-assisted draft saved roughly three days on the documentation phase."*
   Explanation: A specific retro discussion, even one, shows the practice is becoming routine.
5. **Would a new project manager joining the company know to include this disclosure without being told twice?**
   Example: *A new PM follows the template and fills in the AI section unprompted.* **Explanation:** Default compliance without repeated reminders shows the practice has taken hold.

Stage 4 — Integrated



Every project template includes AI use disclosure, risk notes, and acceptance criteria accounting for AI-assisted work.

Key questions to consider

1. **Does every active project's template include AI disclosure, risk notes, and adjusted acceptance criteria?**
   Example: *All five current projects use the updated template with all three AI-related sections completed.*
   Explanation: Universal, consistent use across all live projects is the Stage 4 bar.
2. **Are acceptance criteria explicitly written to account for AI-assisted work?**
   Example: *A deliverable's acceptance criteria states, "AI-drafted sections must be human-reviewed before sign-off."* **Explanation:** Acceptance criteria reflecting AI-aware standards, not just disclosure, marks deeper integration.
3. **Is there a running comparison of AI-assisted vs. non-AI-assisted project components' rework rates?**
   Example: *A tracked spreadsheet shows AI-assisted sections have a 5% rework rate versus 12% for manual sections.* **Explanation:** An ongoing comparative metric shows the organization is genuinely measuring impact, not just logging use.
4. **Do project retrospectives routinely, not occasionally, include an AI-impact discussion?**
   Example: *Every retro template has a standing "AI impact" agenda item.* **Explanation:** Routine inclusion, versus a one-off mention, is what "integrated" requires.
5. **Would project estimation ever be adjusted based on planned AI use?**
   Example: *A quote to a client factors in a shorter timeline because AI-assisted drafting is planned.* **Explanation:** AI factoring into estimation itself shows it has become structurally embedded in project planning.

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Stage 5 — Optimizing



Retrospectives quantify AI's contribution to schedule, cost, or quality; data feeds future estimation.

Key questions to consider

1. **Does historical AI-impact data feed directly into how future projects are estimated?**
   Example: *The estimation spreadsheet has a formula adjustment based on two years of AI-assisted project data.*
   Explanation: Data directly informing future estimates is the clearest Stage 5 signal.
2. **Has a project retrospective ever led to a company-wide process change?**
   Example: *A retro finding about AI-assisted testing led to a permanent change in the standard project template.*
   Explanation: Retro insights propagating into standing process is a mark of continuous improvement.
3. **Can the organization quantify, across projects, how much AI has moved cost, schedule, or quality?**
   Example: *Leadership can state, "AI-assisted projects finish 15% faster on average," with real data behind it.*
   Explanation: An aggregate, cross-project number is strong evidence of mature, optimized practice.
4. **Are project risk assessments for AI use proactive rather than reactive?**
   Example: *New project risk templates already include emerging risks like AI hallucination in client-facing content, before any incident occurred.* **Explanation:** Proactive risk anticipation, not just incident response, shows real maturity.
5. **Would clients or partners view this organization's AI-aware project management as a competitive differentiator?**
   Example: *A prospective client cites the company's transparent AI disclosure practices as a reason they chose to work with them.* **Explanation:** External market recognition of the practice is a strong, outward-facing proof point of Stage 5.

What It Looks Like in Practice

Milestone checklist

1. Project template updated to disclose AI use
2. Accountable human reviewer named per project
3. Retrospective captures AI impact

KPIs (quantitative)

1. % of projects using the updated template
2. Average project cycle time
3. Defect/rework rate on AI-assisted vs. non-AI-assisted project components

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When Things Don’t Go as Planned

If a disclosure template isn't in place yet for a project already underway, don't wait for the next project cycle to start being honest — retrofit a lightweight disclosure note now for what's already shipped, and adopt the full template starting with the next deliverable.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study, the project template goes from zero mention of AI to a required disclosure field, and eventually to two years of data feeding directly into how new projects are quoted.

> *“AI is not a risk-free tool. But by setting clear guidelines for your team, you can reduce or mitigate certain risks like data privacy issues, bias or using AI-generated content without checking for hallucinations.”*
> — Wipfli — AI Governance Checklist for Mid-Market Companies (2026); the logic behind disclosing AI use and naming an accountable reviewer per project

Case study: Mylar Technologies (illustrative composite)

Mylar Technologies is a fictional small scale software company building custom scheduling and workflow integrations for mid-market clients, used throughout this framework as a running, illustrative example. It is a composite, not a verified real-world case, built to show how one organization's practice might look at each stage.

Stage	Mylar Technologies at this stage
1 Aware	The current ticketing-dashboard project plan has zero mention of AI, and no one considers it a risk category.
2 Experimenting	One project informally uses an AI tool to draft the requirements doc; the lead mentions in a retro that it "saved some time."
3 Adopting	The updated project template adds a required "AI tools used" field, and the tech lead is named the accountable reviewer for AI-assisted deliverables.
4 Integrated	All five active projects use the new template with AI disclosure, risk notes, and adjusted acceptance criteria, and rework rates on AI-assisted sections are tracked against manual ones.
5 Optimizing	Two years of cross-project data now feed directly into how Mylar estimates new projects, and a prospective client cites the company's transparent AI practices as a reason they chose to work with them.

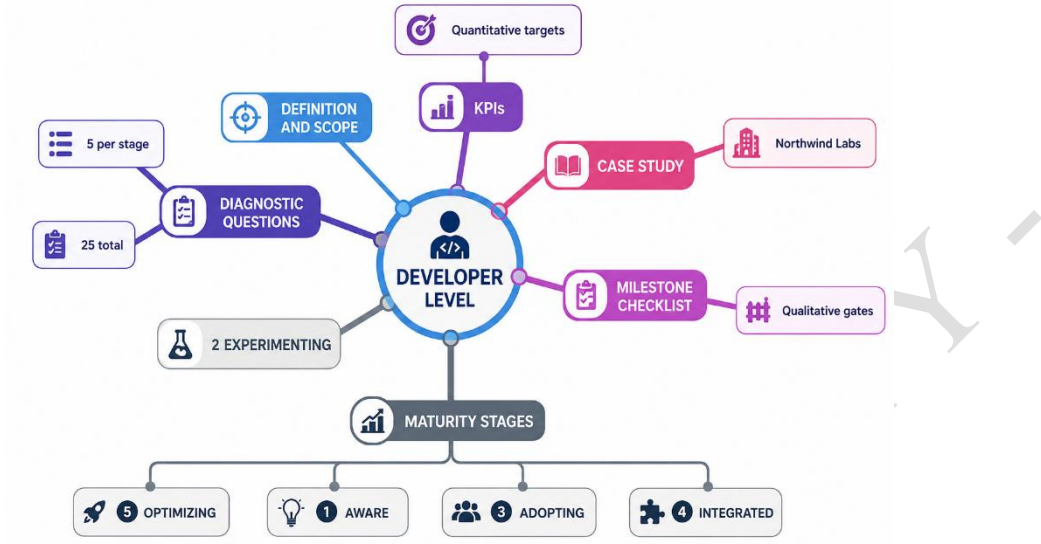
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What You've Achieved

You'll know project-level adoption has matured when estimating a new project's timeline automatically factors in planned AI use, the same way it would factor in team size or scope.

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Chapter 15: Developer Level



Covers individual and team use of AI coding assistants for writing, reviewing, and documenting code.

Context from the current tool landscape (2026): roughly 90% of professional developers now use at least one AI coding tool regularly, and 74% have adopted a specialized assistant beyond a general chatbot. GitHub Copilot remains the most broadly adopted tool, especially in larger companies, while Claude Code has grown rapidly (from roughly 3% to 18–24% adoption within about a year) and currently leads on developer satisfaction. Cursor holds a similar adoption share to Claude Code. A small organization does not need to chase every new entrant — it needs one or two approved, well-integrated tools and a clear code-review standard for AI-generated code.

Setting the Scene

Developer-level adoption is where AI's productivity promise meets the codebase directly — and where, per the JetBrains AI Pulse survey (January 2026), 90% of developers already regularly use at least one AI tool at work, whether or not their organization has caught up with a policy.

Where Most Teams Start

Most developers start exactly the way Mylar Technologies' backend developer did: a personal coding-assistant subscription the company doesn't know about, no review policy for AI-generated code, and zero visibility for engineering leadership into who's using what.

A Better Approach

The better approach brings that usage into the open deliberately — an approved tool, an explicit code-review rule for AI-generated code, and tracked usage — so the productivity gain is governed rather than accidental.

Foundational Concepts

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The Developer level is where AI adoption is furthest along industry-wide, and where the tooling landscape moves fastest. It covers how developers write, review, and document code with AI assistance.

What This Section Helps You Achieve

The goal is a team where an approved coding assistant is standard practice, AI-generated code gets the same review bar as human-written code, and the team can point to real numbers — not just a feeling — that adoption is working.

The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

Step by Step — The Five Maturity Stages

The rest of this section works through each maturity stage in depth: a short description, then five diagnostic questions with an example and explanation for each, to help you pin down exactly which stage you're in. Each stage opens with the same five-stage roadmap, with the current stage highlighted, so it's always clear where a given stage sits relative to the whole scale.

Stage 1 — Aware



Developers may use AI tools personally/unofficially; no company policy.

Key questions to consider

1. **Are developers using personal AI coding tools without any company visibility?**
   Example: *A developer has a personal coding-assistant subscription paid out of pocket, unknown to the company.* **Explanation:** Shadow, unofficial use with zero visibility is the Stage 1 signature.
2. **Is there any company policy — even one sentence — about AI-generated code?**
   Example: *The engineering handbook says nothing about AI at all.* **Explanation:** Total policy silence confirms no governance yet.
3. **Do code reviewers know whether a given pull request contains AI-generated code?**
   Example: *A reviewer has no way of knowing if a PR was AI-assisted or fully hand-written.* **Explanation:** Zero visibility into AI's role in code review is typical of Stage 1.
4. **Has engineering leadership ever discussed AI coding tools as a team topic?**
   Example: *AI tools have never come up in an engineering all-hands.* **Explanation:** Absence from leadership conversation confirms the topic hasn't landed yet.
5. **If a developer left tomorrow, would anyone know which AI tools they'd been relying on?**
   Example: *No one tracks individual tool usage at all.* **Explanation:** No visibility into individual practice reflects an ungoverned, pre-Stage-2 state.

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Stage 2 — Experimenting



Company approves a pilot tool for a subset of developers.

Key questions to consider

1. **Has the company officially approved at least one AI coding tool for a subset of developers to pilot?**
   Example: *Five backend developers get sanctioned trial licenses for an AI coding assistant.* **Explanation:** An official, even small, pilot is the entry point to Stage 2.
2. **Do the pilot developers have a way to report back what's working?**
   Example: *The pilot group posts weekly notes in a shared doc about the tool's strengths and weaknesses.*
   Explanation: A feedback loop, even informal, marks meaningful experimentation.
3. **Has the pilot surfaced a concrete concern that leadership is aware of?**
   Example: *A pilot developer flags that the assistant sometimes suggests deprecated library calls.* **Explanation:** Surfacing real, specific concerns, not just vague worry, shows genuine hands-on experimentation.
4. **Is there rough awareness of how much the pilot group actually uses the tool?**
   Example: *Leadership knows the pilot group uses it "most days" without hard numbers.* **Explanation:** Rough, informal usage awareness is appropriate for this stage.
5. **Could the pilot be stopped tomorrow without much disruption?**
   Example: *If licenses were revoked, the five developers would just go back to their old workflow.* **Explanation:** Low dependency confirms the practice hasn't become load-bearing yet.

Stage 3 — Adopting



An approved coding assistant is rolled out to the whole engineering team, with a code-review rule for AI-generated code.

Key questions to consider

1. **Is an approved coding assistant rolled out to the whole engineering team, not just a pilot group?**
   Example: *All twelve developers now have licensed seats for the approved tool.* **Explanation:** Full-team rollout is the clear Stage 3 threshold.
2. **Is there an explicit code review rule stating AI-generated code gets the same scrutiny as human-written code?**
   Example: *The PR template includes a checkbox: "Contains AI-assisted code — reviewed to the same standard."*
   Explanation: An explicit, applied review rule is central to responsible Stage 3 adoption.

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3. **Do most developers, not just early adopters, use the approved tool weekly?**
   Example: *Ten of twelve developers show active weekly usage in the tool's admin dashboard.* **Explanation:** Majority regular use, verifiable via data, distinguishes Stage 3 from Stage 2.
4. **Has anyone tracked a real before/after number on cycle time or defect rate?**
   Example: *Average PR turnaround dropped from two days to 1.5 days after rollout, per the team's tracker.* **Explanation:** A real, tracked metric, even one, shows evidence-based adoption has begun.
5. **Would a new developer be trained on the approved tool and review rule during onboarding?**
   Example: *Onboarding now includes a session on the approved AI coding assistant and the review policy.* **Explanation:** Onboarding inclusion shows the practice has become a default, not an opt-in.

Stage 4 — Integrated



AI-assisted development is standard; metrics tracked and compared pre/post adoption; developers trained on prompt practices.

Key questions to consider

1. **Are velocity and defect-rate metrics routinely compared pre/post AI adoption, not just once?**
   Example: *A quarterly engineering dashboard tracks defect-rate trends since AI tool rollout.* **Explanation:** Ongoing, routine comparison, not a one-time study, is the Stage 4 bar.
2. **Are developers trained on prompt practices and known failure modes, not just given tool access?**
   Example: *An internal wiki page documents common AI coding pitfalls and how to catch them.* **Explanation:** Structured training on failure modes shows mature, embedded practice.
3. **Is AI-assisted development the default expectation, with manual-only coding the exception requiring justification?**
   Example: *A developer who prefers not to use the assistant needs to explain why in a retro.* **Explanation:** Flipping the default so AI-assisted is normal marks full integration.
4. **Do code review comments about AI-generated code issues show a downward trend over time?**
   Example: *The team tracks that AI-related review comments dropped 40% over two quarters as practice matured.* **Explanation:** A declining trend in AI-specific review friction shows the team has learned to use the tool well, not just adopted it.
5. **Is tool selection revisited periodically based on data, not locked in indefinitely?**
   Example: *Engineering leadership re-evaluates the tool choice every two quarters against new entrants and performance data.* **Explanation:** Periodic, evidence-based re-evaluation shows the practice is actively managed, not "set and forget."

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Stage 5 — Optimizing



Developers use AI agents for larger units of work under strong review discipline; workflows and prompts continuously tuned.

Key questions to consider

1. **Are developers using AI agents for larger units of work, not just autocomplete?**
   Example: *A developer delegates an entire small feature — code, tests, and docs — to an AI agent under review.*
   Explanation: Agentic, larger-scope use, versus line-by-line suggestions, is the clearest Stage 5 marker.
2. **Does the team have a habit of tuning prompts/workflows based on what's been learned?**
   Example: *The team maintains and updates a shared prompt playbook monthly based on what's worked.*
   Explanation: Active, ongoing tuning, not "set once," shows continuous optimization.
3. **Are learnings from this team's AI practice shared and adopted by other teams in the company?**
   Example: *This team's code-review checklist for AI-generated code becomes the company standard.* **Explanation:** Org-wide adoption of this team's practice is strong evidence of leading-edge maturity.
4. **Can the team point to a measurable business outcome, not just developer convenience, tied to its AI practice?**
   Example: *Faster feature delivery attributable to AI-assisted development helped ship a feature ahead of a competitor.* **Explanation:** Business-level impact, not just developer happiness, is the top-tier proof point.
5. **Would the team's AI practice survive turnover of its most senior/enthusiastic member?**
   Example: *The practice is documented and habitual enough that if the lead advocate left, the team would keep using it the same way.* **Explanation:** Resilience to individual turnover shows the practice is institutional, not personality-dependent.

What It Looks Like in Practice

Milestone checklist

1. Approved coding assistant(s) named
2. License/seats provisioned
3. Code review policy explicitly covers AI-generated code
4. Developers onboarded/trained
5. Usage tracked

KPIs (quantitative)

1. % of developers actively using an approved AI tool weekly
2. Change in cycle time per PR/ticket
3. Change in defect/bug rate post-adoption
4. % of code review comments related to AI-generated code issues (should trend down as practice matures)

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When Things Don’t Go as Planned

If engineering leadership isn't ready to officially approve a tool yet, don't let shadow usage continue ungoverned in the meantime — at minimum, add the one-line PR checkbox ("contains AI-assisted code, reviewed to the same standard") immediately, and revisit formal tool approval within the quarter.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study, engineering goes from one unofficial personal subscription to delegating whole features to AI agents under review.

> *“90% of developers regularly used at least one AI tool at work for coding and development tasks.”*
> — JetBrains AI Pulse survey, January 2026
>
> *“The shift toward best-of-breed agents demonstrates that product excellence now outweighs ecosystem lock-in. When a standalone tool offers clear superiority, it renders integrated stacks obsolete; developers will always migrate to the individual components that actually deliver the best results.”*
> — Mikhail Bogdanov, JetBrains (2026)

Case study: Mylar Technologies (illustrative composite)

Mylar Technologies is a fictional small scale software company building custom scheduling and workflow integrations for mid-market clients, used throughout this framework as a running, illustrative example. It is a composite, not a verified real-world case, built to show how one organization's practice might look at each stage.

Stage	Mylar Technologies at this stage
1 Aware	A backend developer has a personal coding-assistant subscription the company doesn't know about, and the engineering handbook says nothing about AI.
2 Experimenting	Mylar officially approves a pilot for five developers, who post weekly notes about what's working.
3 Adopting	All twelve developers get licensed seats, and the PR template adds a checkbox confirming AI-assisted code was reviewed to the same standard as human code.
4 Integrated	A quarterly dashboard tracks defect rate and cycle-time trends since rollout, and a wiki page documents common AI coding pitfalls the team has learned to catch.
5 Optimizing	Developers now delegate whole small features — code, tests, and docs — to an AI agent under review, and faster delivery attributable to this practice helped Mylar ship a feature ahead of a competitor.

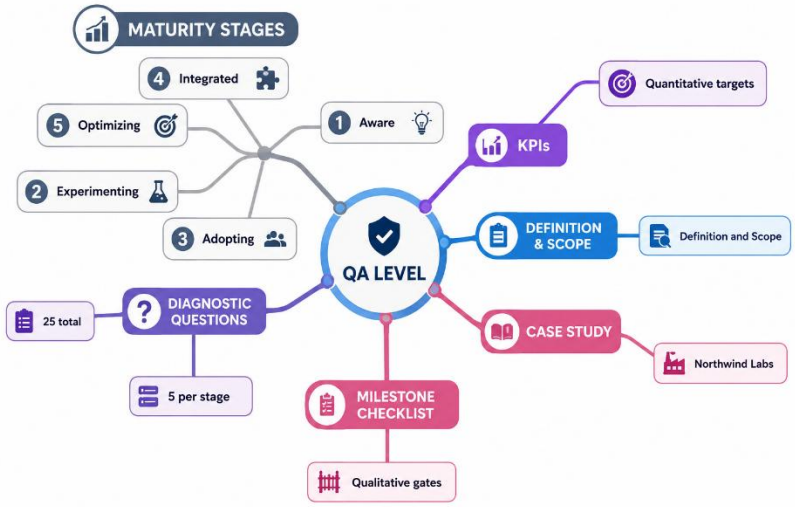
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What You've Achieved

You'll know developer-level adoption has matured when AI-assisted work is the default and the exception — a developer choosing not to use it — is the thing that needs explaining, not the other way around.

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Chapter 16: QA Level



Covers use of AI in test generation, maintenance, execution, and defect triage.

Context: a widely used 2026 AI-QA maturity model describes five stages from manual QA through fully autonomous quality engineering, and ties each stage to concrete numbers — for example, at the "AI-Assisted" stage, test-suite maintenance should consume under 10% of QA engineer time, and at the "AI-Native" stage, mean time to diagnose a failing test should fall under 30 minutes. A funded small organization adopting AI QA tooling should realistically target reaching the "AI-Assisted/Integrated" stage within about six months.

Setting the Scene

QA is where AI adoption gets measured most precisely, because testing produces hard numbers almost by default — pass rates, coverage, defect counts — and a 2026 AI-QA maturity model ties each stage to concrete benchmarks, like test-suite maintenance dropping under 10% of QA capacity at the "AI-Assisted" stage.

Where Most Teams Start

Most QA teams start fully manual — hand-scripted testing, no automation, releases regularly delayed while a QA engineer clicks through the app by hand — exactly as Mylar Technologies did before its two-day manual release cycles became the problem worth solving.

A Better Approach

The better approach treats QA maturity as a measurable progression, not a one-time tool purchase — tracking maintenance time, diagnosis time, and coverage breadth as the AI-assisted practice takes hold, so QA becomes a velocity multiplier rather than the bottleneck that delays every release.

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Foundational Concepts

The QA level is where AI adoption is measured most precisely, because testing produces hard numbers almost by default — pass rates, coverage, defect counts. It covers how AI assists with generating, maintaining, and triaging tests.

What This Section Helps You Achieve

The goal is for QA to stop being the bottleneck that delays every release and become, in the industry's own words, a multiplier on engineering velocity instead.

The stage-by-stage detail for how this level actually moves through the five maturity stages — a description, five diagnostic questions with examples and explanations, and a Mylar Technologies snapshot for each stage — is covered in depth in "Detailed stage-by-stage diagnostics" below, rather than repeated here.

Step by Step — The Five Maturity Stages

The rest of this section works through each maturity stage in depth: a short description, then five diagnostic questions with an example and explanation for each, to help you pin down exactly which stage you're in. Each stage opens with the same five-stage roadmap, with the current stage highlighted, so it's always clear where a given stage sits relative to the whole scale.

Stage 1 — Aware



Manual, hand-scripted testing; no automation; releases are frequently delayed by QA.

Key questions to consider

1. **Is testing primarily manual and hand-scripted, with no automated suite at all?**
   Example: *Every release requires a QA engineer to click through the app manually for two days.* **Explanation:** Fully manual testing with no automation is the Stage 1 baseline.
2. **Do releases regularly slip because of QA bottlenecks?**
   Example: *The last three releases were each delayed over a week waiting on manual test cycles.* **Explanation:** Chronic QA-caused delays are a classic symptom of Stage 1.
3. **Has anyone on the team discussed using AI for testing, even hypothetically?**
   Example: *No one has brought up AI as a possible solution to the QA bottleneck.* **Explanation:** Not even considering the option shows the team hasn't engaged with it yet.
4. **Would the QA team know where to start if asked to explore AI testing tools tomorrow?**
   Example: *If asked, the QA lead would have no idea which tools even exist.* **Explanation:** Lack of basic orientation confirms Stage 1.
5. **Is test coverage limited mostly to a handful of critical happy-path scenarios?**

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Example: *Only the core checkout flow is tested; edge cases are untested.* **Explanation:** Minimal, ad hoc coverage is typical when testing is entirely manual and time-constrained.

Stage 2 — Experimenting



Automated tests exist but are fragile; coverage limited to happy paths; pass rate below 90% on stable code.

Key questions to consider

1. **Does an automated test suite exist, even if it's fragile?**
   Example: *An automated suite covers the login and checkout flows but breaks on minor UI changes.* **Explanation:** Existence of automation, however fragile, is the Stage 2 entry point.
2. **Do tests generate frequent false failures that require manual triage?**
   Example: *Roughly a third of failed test runs turn out to be flaky, not real bugs.* **Explanation:** Frequent false failures are the hallmark of early, unstable automation.
3. **Is test pass rate on stable code below 90%?**
   Example: *The suite passes only 85% of the time even when no real bugs are introduced.* **Explanation:** A low, unreliable pass rate on stable code indicates immature automation, per this stage's benchmark.
4. **Is coverage limited mostly to happy-path scenarios, now automated instead of manual?**
   Example: *The automated suite covers the same narrow set of flows that used to be tested manually.*
   Explanation: Automating the same narrow scope without expanding it is typical of this transitional stage.
5. **Has anyone floated the idea of using AI to fix the flakiness problem?**
   Example: *A QA engineer suggests trying an AI-based self-healing tool after yet another flaky failure.* **Explanation:** The problem becoming acute enough to prompt this suggestion is a natural bridge to Stage 3.

Stage 3 — Adopting



AI tools augment workflows through self-healing selectors and test generation; maintenance overhead starts dropping.

Key questions to consider

1. **Has an AI tool for test generation or self-healing selectors been introduced?**

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Example: *The team adopts an AI tool that automatically updates broken selectors when the UI changes.*
Explanation: Introducing a specific AI tool for a specific pain point is the Stage 3 marker.

2. **Has test-suite maintenance time started dropping since introducing the tool?**
   Example: *Maintenance time drops from roughly 25% of QA capacity to 15% within the first month.* **Explanation:** A measurable drop in maintenance overhead is the expected, trackable early win.
3. **Is the drop in maintenance time actually being measured and reported, not just assumed?**
   Example: *The QA lead tracks maintenance hours in a spreadsheet before and after the tool's introduction.*
   Explanation: Real measurement, versus a vague sense that things feel better, distinguishes genuine adoption.
4. **Is coverage starting to expand beyond the original happy-path scenarios?**
   Example: *The team adds edge-case tests that would have been too costly to maintain manually before.*
   Explanation: Expanding scope, enabled by lower maintenance cost, is a natural sign of Stage 3 progress.
5. **Is the defect escape rate being tracked, even if it hasn't improved dramatically yet?**
   Example: *The team starts logging how many bugs reach production versus getting caught pre-release.*
   Explanation: Establishing the baseline metric is a prerequisite for later, more dramatic improvement.

Stage 4 — Integrated



AI embeds throughout the lifecycle; QA is a multiplier on engineering velocity; mean diagnosis time under 30 minutes.

Key questions to consider

1. **Is AI embedded across the QA lifecycle, not just one point tool?**
   Example: *AI now assists with test generation, visual diffing, and automatically categorizing failure causes.*
   Explanation: Multi-point embedding across the lifecycle is the Stage 4 bar, versus a single point solution.
2. **Is test-suite maintenance time reliably under 10% of QA engineer capacity?**
   Example: *The team's tracked maintenance time has stabilized at 8% for two consecutive quarters.* **Explanation:** Meeting this specific, industry-referenced benchmark is a concrete Stage 4 marker.
3. **Is mean time to diagnose a failing test consistently under 30 minutes?**
   Example: *The failure-triage dashboard shows an average diagnosis time of 22 minutes.* **Explanation:** Hitting this benchmark reliably, not just occasionally, confirms integration, not just a good week.
4. **Would engineering describe QA as a velocity multiplier rather than a bottleneck?**
   Example: *In a retro, engineers note that QA no longer holds up releases and instead catches issues early enough to speed things up.* **Explanation:** This qualitative shift in how QA is perceived is a strong, if informal, integration signal.
5. **Is coverage breadth tracked and trending up week over week?**
   Example: *A coverage dashboard shows steady growth over the last two quarters.* **Explanation:** Sustained, visible growth, not a one-time jump, reflects an integrated, ongoing practice.

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Stage 5 — Optimizing



Substantially self-managing pipelines generate, maintain, and prioritize tests autonomously; coverage grows without headcount growth.

Key questions to consider

1. **Does the pipeline largely self-manage — generating, maintaining, and prioritizing tests with minimal manual intervention?**

   Example: *New feature branches automatically get relevant AI-generated tests without a human writing them from scratch.* **Explanation:** Substantial self-management, not just AI assistance, is the Stage 5 threshold.

2. **Is coverage growing without a proportional increase in QA headcount?**

   Example: *Test coverage doubled over a year while the QA team stayed the same size.* **Explanation:** Coverage growth decoupled from headcount is strong evidence of a scalable, optimized system.

3. **Is CI pipeline run time actively reduced through predictive test selection, not just brute-force parallelization?**

   Example: *The CI system intelligently runs only the tests most likely affected by a given change, cutting run time significantly.* **Explanation:** Smart, predictive optimization, versus just throwing more compute at it, shows real sophistication.

4. **Does the QA function proactively flag quality risks before they're requested, rather than reactively responding to incidents?**

   Example: *The system flags a risky pattern in a new pull request before any human asked it to look.* **Explanation:** Proactive, autonomous risk-flagging is characteristic of a mature, self-managing pipeline.

5. **Has this QA practice become something other companies or teams specifically want to learn from?**

   Example: *A partner engineering team asks to shadow this QA process as a model for their own transformation.*
   Explanation: External recognition as a reference model is a strong outward proof point of Stage 5 maturity.

What It Looks Like in Practice

Milestone checklist

1. AI test-generation/self-healing tool selected and approved
2. Pilot suite converted
3. Maintenance time measured before/after
4. Coverage tracked over time

KPIs (quantitative)

1. Defect escape rate (trending down)
2. Test-suite maintenance time as % of QA capacity (target under 10% by Stage 3–4)
3. Mean time to diagnose a failing test (target under 30 minutes by Stage 4)

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4. Test coverage breadth (trending up week over week)
5. CI pipeline run time

When Things Don’t Go as Planned

If mutation testing or perceptual-diffing tooling isn't available yet to confirm AI-generated tests are real, don't skip verification — fall back to a second QA engineer manually reviewing test names, assertions, and data before trusting a generated suite, and revisit the tooling gap once budget allows.

Scenarios and Use Cases

Case study and references

In the Mylar Technologies case study, QA goes from two-day manual release cycles to a partner company asking to shadow its process as a model for their own transformation.

> *“AI in QA testing is not a tool you buy, it is a set of capabilities you sequence.”*
> — remote.qa — AI QA Testing Guide (2026)
>
> *“Test suite maintenance takes less than 10% of QA engineer time.”*
> — remote.qa — Level 3 indicator, AI QA Testing Guide (2026)
>
> *“Mean time to diagnosis after a failure is under 30 minutes.”*
> — remote.qa — Level 4 indicator, AI QA Testing Guide (2026)

Case study: Mylar Technologies (illustrative composite)

Mylar Technologies is a fictional small scale software company building custom scheduling and workflow integrations for mid-market clients, used throughout this framework as a running, illustrative example. It is a composite, not a verified real-world case, built to show how one organization's practice might look at each stage.

Stage	Mylar Technologies at this stage
1 Aware	Every release requires two days of manual click-through testing, and releases have slipped three times in a row.
2 Experimenting	An automated suite exists but breaks on minor UI changes, with about a third of failures turning out to be flaky rather than real bugs.
3 Adopting	The team adopts an AI self-healing tool for selectors, and maintenance time drops from roughly 25% of QA capacity to 15% within the first month.
4 Integrated	Maintenance time has stabilized at 8% for two straight quarters, mean diagnosis time for failing tests averages 22 minutes, and engineering now describes QA as a velocity multiplier rather than a bottleneck.
5 Optimizing	New feature branches automatically get AI-generated tests without anyone writing them from scratch, and a partner company's engineering team asks to shadow Mylar's QA process as a model for their own transformation.

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What You've Achieved

You'll know QA has matured when test-suite maintenance reliably costs under 10% of QA capacity, a failing test gets diagnosed in under 30 minutes, and engineering describes QA as something that speeds releases up rather than holding them back.

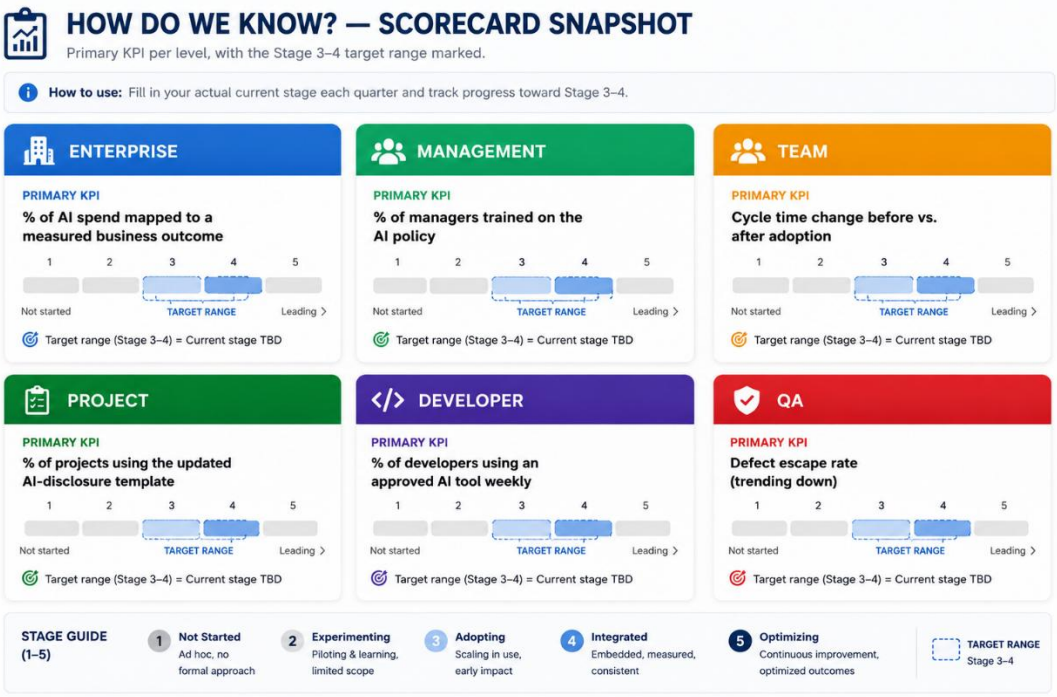
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Chapter 17: Measuring Success — the "How Do We Know" Scorecard

Success is confirmed with three combined instruments, reviewed together on a quarterly cadence:

1. **Maturity scorecard:** each of the six levels is rated 1–5 against the tables above, using the diagnostic questions as the practical test. This produces a single-page snapshot of where the organization stands.
2. **KPI dashboard:** the quantitative metrics listed per level are tracked over time (a simple spreadsheet is sufficient at this size).
3. **Milestone checklist audit:** the qualitative checklist items per level are marked done/not done; incomplete items become the next quarter's priorities.

Sample scorecard



Level	Current Stage (1-5)	Target Stage	Primary KPI to watch
Enterprise	—	—	% AI spend mapped to measured outcome
Management	—	—	% managers trained on AI policy
Team	—	—	Cycle time change
Project	—	—	% projects on updated template
Developer	—	—	% developers using approved tool weekly
QA	—	—	Defect escape rate

A level only "counts" as reaching a stage once both the checklist items and the KPI targets for that stage are met — a checklist without moving numbers, or numbers without documented practice, both signal the stage isn't real yet.

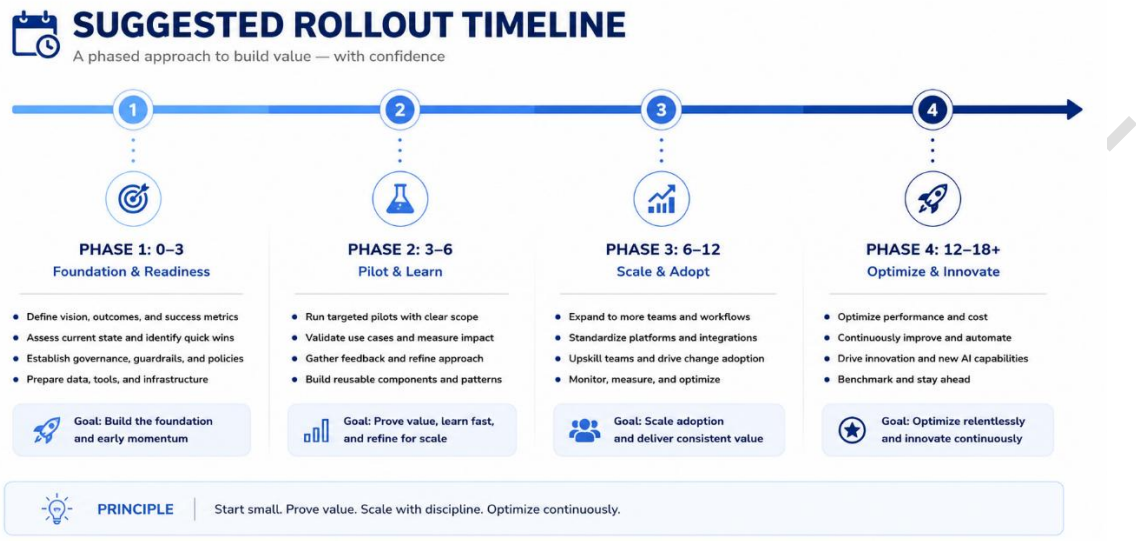
Illustrative maturity heatmap

A heatmap view makes uneven progress across levels easy to spot at a glance — which levels are ahead, which are on pace, and which need attention this quarter. The snapshot below is the Mylar Technologies composite, 18 months into a deliberate rollout.

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Chapter 18: Suggested Rollout Timeline

With governance and diagnostics in place, rollout is sequenced rather than simultaneous — foundational levels first, technical levels scaled once the foundation holds.



1. **Months 0–3:** Enterprise and Management foundations — name the AI lead, write the policy, approve the tool list, train managers.
2. **Months 3–6:** Team and Project level rollout — pick pilot teams, update project templates, start the knowledge-sharing habit.
3. **Months 6–12:** Developer and QA level scaling — roll out approved coding assistant and AI QA tooling org-wide, start tracking the KPIs above, target Stage 3–4.
4. **Beyond 12 months:** push toward Stage 5 (Optimizing) in Developer and QA levels first, since tooling there is most mature; keep Enterprise/Management/Team/Project at a sustained Stage 4.

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Chapter 19: Capability Matrix Across Organization Sizes

This book's core numbered framework is scoped for organizations under 100 employees, but the underlying discipline scales. The matrix below is illustrative guidance — reasoned from the same frameworks used throughout the book, not a verified benchmark study — for calibrating expectations across a wider range of organization sizes.

Org. Size	Maturity Target	Core Capability First	Investment Pattern	ROI Timeframe	Governance Effort	Tech. Complexity
Startup (<20 people)	Stage 2-3	One AI lead wearing multiple hats; a single approved tool	Low — mostly per-seat SaaS licensing	Weeks to a few months on a single pilot	Minimal — a one-page policy is enough	Low — embedded / BYOAI, no custom build
Small (20-100)	Stage 3-4	Named AI lead, written policy, one or two pilot teams	Low to moderate — licensing plus limited integration	One to two quarters per top opportunity	Light — the governance foundation in Part II §3	Low to moderate — mostly BYOAI, light RAG
Mid-Market (100-1,000)	Stage 3-4, Stage 4-5 in Developer/QA	Cross-functional governance committee; per-level rollout owners	Moderate — dedicated budget line, some custom integration	Two to four quarters, tracked per initiative	Moderate — governance committee plus per-project review	Moderate — RAG and workflow tooling common
Enterprise (1,000-10,000)	Stage 4 target org-wide, Stage 5 in pockets	Formal AI operating model (see Role Responsibility Matrix); platform team	Significant — platform investment, dedicated AI engineering	Ongoing portfolio view, not single-project ROI	Substantial — formal risk review, vendor governance, audit trail	Moderate to high — mixed build/buy, some fine-tuning
Large Enterprise (10,000+)	Stage 4-5 (strategic edge)	Enterprise-wide platform, agentic AI governance, model risk management	Large — often a dedicated AI/ML organization	Measured as a standing capability, not a project	Extensive — formal committees, regulatory alignment, third-party audit	High — custom architectures, fine-tuning, agentic systems at scale

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Chapter 20: Top AI Transformation Mistakes

Readers often learn faster from what goes wrong than from what goes right, and this space now has enough track record to name its recurring mistakes plainly. The eight below are organizational and strategic-level anti-patterns — the ones that show up in how a transformation is run, not in a specific line of generated code. Chapter 21 already catalogs the technical failure modes (Hallucination Spiral, Comprehension Debt, and the like) that show up once code is being written; this chapter doesn't repeat those and cross-references them where relevant.

Shadow AI

Employees adopt AI tools on their own, unofficially, because no sanctioned option exists — data ends up in unvetted tools and nobody is accountable for the output.

> **FIX:** Publish an approved tool allow-list within the first week of the effort (Chapter 10), even before a full policy is written. A short, explicit "yes list" beats a long, unenforced "no."

Tool-first thinking

The initiative starts with "which AI tool should we buy" instead of "which real problem are we solving." This is how organizations end up with an expensive license and no adoption.

> **FIX:** Run opportunity identification (Chapter 3) before evaluating any vendor. The tool choice is the last decision in Part I, not the first.

No governance

Adoption scales past the pilot with no accountable owner and no policy, on the theory that governance can be added later once things are moving. It rarely gets added later — it gets added after an incident.

> **FIX:** The governance foundation (Chapter 10) is designed to take days, not months, precisely so there's no excuse to defer it.

Measuring activity instead of value

Dashboards track logins, prompts sent, or "AI-assisted" tickets closed — all activity metrics — without ever connecting to a business outcome leadership actually cares about.

> **FIX:** Every KPI in Chapters 11–17 and the ROI frame in Chapter 27 is chosen because it ties to a real outcome (cost, cycle time, defect rate), not because it's easy to log.

Poor prompt discipline

Context is packed in ad hoc, instructions and general conventions are mixed together, and nobody owns a shared standard — every developer or analyst reinvents their own approach.

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FIX: Chapter 21's context-and-prompt-engineering standards separate instructions from guidance and centralize both in a shared rules file, rather than leaving it to individual habit.

Missing human review

AI-generated output — code, analysis, customer-facing text — ships without anyone checking it, on the assumption that the model is reliable enough not to need a second look.

FIX: A named accountable reviewer is required at every level in this book, from Part II's per-project reviewer to Chapter 21's non-negotiable CI review gates. AI is never the accountable party (Principle 4).

Pilot purgatory

A pilot proves the concept, gets positive anecdotal feedback, and then never graduates — it just runs indefinitely without a decision to scale, kill, or formally adopt it, quietly consuming attention and credibility.

FIX: Every pilot needs a defined decision gate at a specific checkpoint (Chapter 5's business case, and the Pilot stage of the Transformation Journey) — continue, adjust, or kill, not drift.

Confusing a vendor's success story with your own

Leadership sets expectations based on a vendor's best-case marketing case study, then is blindsided when internal results land far short — the same mistake Chapter 28 was built to prevent.

FIX: Weight vendor-published case studies as illustrative, not predictive, and look for independent or peer-reviewed evidence before setting an internal target (Principle 9).

CHAPTER RECAP

1. Eight recurring anti-patterns show up above and before the code: Shadow AI, tool-first thinking, no governance, measuring activity instead of value, poor prompt discipline, missing human review, pilot purgatory, and confusing a vendor's story with your own.
2. Chapter 21 catalogs the technical failure modes that show up once code is being written (Hallucination Spiral, Comprehension Debt, and the like) — this chapter deliberately does not repeat those.

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PART III : AI Execution Playbook

Practical, decision-maker-level depth on the topics that matter once adoption is underway: developer and QA practice beyond the basics, agentic AI, RAG, MCP, local vs. cloud models, ROI measurement, and six real, named, credibility-labeled enterprise case studies.

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Chapter 21: Developer Practices — Beyond Basic Adoption

Setting the Scene

AI coding assistants are now mainstream in engineering orgs, but adoption maturity varies wildly team to team. Getting a tool approved and into daily use is only the first milestone; what happens next — how context is managed, how output is reviewed, how security holds up at machine speed — determines whether AI coding delivers real throughput or just moves the risk downstream.

Where Most Teams Start

Most teams treat tool approval as the finish line: the assistant is approved, developers are using it day to day, and it looks like the adoption problem is solved.

Chapter 15 (Developer Level) covers getting an approved coding assistant into regular use. This section is for teams that have already done that and are past the point of just "using the tool" — where the real work is context discipline, review structure, and avoiding failure modes that only show up at scale.

1. **Context and prompt engineering standards:** Separate instructions (direct directives) from guidance (general conventions the model should always follow), and build context gradually from real usage rather than guessing upfront — excess context degrades results and raises cost. Practically: plan first with a short spec before generating code, pack context deliberately, commit frequently so there's always a rollback point, and encode your team's standards in a rules file (an agents.md / CLAUDE.md-style file) that lives alongside your linter config rather than replacing it.
2. **Code review for AI-generated code:** Tier review depth by risk, not by who (or what) wrote the code. Require evidence before review even starts — a purpose statement, a small diff, test output. Keep AI-generated pull requests small deliberately (agent-authored PRs run roughly 51% larger than human ones by default, which erodes review quality). Scrutinize test-file changes specifically for assertions quietly rewritten to match broken behavior instead of fixing it. Keep CI gates (lint, coverage, security) non-negotiable. The human still owns the merge decision, always.
3. **Security review:** AI-generated code carries measurably more of certain vulnerability classes — one 2026 industry analysis found roughly 2.74x more cross-site-scripting vulnerabilities and 1.7x more logic errors than human-written code, plus recurring "package hallucination" (inventing dependencies that attackers later register and poison) and hardcoded secrets. Traditional static-analysis tools also perform worse here, throwing false-positive rates above 75% on AI-generated code. Banning AI tools outright doesn't work either — a majority of developers keep using unsanctioned tools anyway when banned. The practical answer is embedding validation directly in the IDE/PR workflow rather than relying on a separate, slower review pass.
4. **Known failure modes to watch for:** A useful checklist drawn from 2026 industry write-ups: Hallucination Spiral (confidently wrong, compounding over turns), Comprehension Debt (code nobody on the team actually understands), Context Window Amnesia (the model forgetting earlier decisions), Automation Bias (trusting output because it looks polished), Security Blindness, Architecture Drift, Testing Illusion (tests that pass without testing anything real), and Productivity Theater — individual output rises while team-level delivery stays flat because review overhead absorbs the gain.
5. **Structuring agentic workflows in CI/CD:** A useful pattern is a dual-path pipeline: a fast, low-friction path for experimentation (agents proposing changes in a non-production environment) and an unchanged, strict path for production (the same review, approval, and audit-logging gates that applied before AI was involved). The two paths should share identical access-control and audit logging, even though their speed differs enormously.

> *"There's classic programming principles that to seasoned engineers are common knowledge... You can tell the LLM to build your application that way."*

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> — Vish Abrams, Chief Architect, Heroku — Stack Overflow Blog (2026)
>
> *“You are the accountable engineer.”*
> — Addy Osmani, "My LLM coding workflow going into 2026" (2026)
>
> *“The bottleneck is no longer how fast you write code, it is how fast a trusted human can be confident in a review.”*
> — Addy Osmani, "Agentic Code Review" (2026)
>
> *“You cannot secure machine-speed code generation using human-speed reactive workflows.”*
> — OX Security, on AI code security practices (2026)

Mylar Technologies: In Practice

Three of Mylar's eight developers were already using personal AI coding subscriptions before the support pilot had a name, let alone a policy. Once the pilot needed real engineering time, that shadow usage became the starting point for a real review standard: a shared rules file, mandatory review of AI-authored diffs, and a size cap on any single AI-generated change, so no one was reviewing a 600-line pull request generated in one prompt.

When Things Don’t Go as Planned

If your team recognizes one or more of the failure modes below — especially Productivity Theater, where individual output rises but delivery doesn't — treat it as a signal to revisit context discipline and review structure rather than adding more tooling on top.

What You've Achieved

With these practices in place, your team has moved past ad-hoc tool use into a disciplined AI-assisted development process: context is managed deliberately, review effort is scaled to risk, security checks run at the speed code is generated, and known failure modes are caught early rather than discovered in production.

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Chapter 22: QA Practices — Beyond Basic Adoption

Where Most Teams Start

Most teams that already have AI test generation running treat a passing suite as proof of quality: the AI drafts the test, it runs green in CI, and everyone moves on without checking whether the assertions would catch a real regression.

A Better Approach

The shift this chapter argues for is treating "passing" as a claim to verify, not a result to trust. That means deliberately breaking code to see if generated tests actually fail, reviewing assertions for what they truly check, and keeping AI on a tight leash — fast in pre-production loops, with the same strict human gates before anything ships.

Chapter 16 (QA Level) covers introducing AI test generation and self-healing selectors. This section is for teams further along, where the risk shifts from "no automation" to a subtler one: AI-generated tests that pass without actually testing anything real.

1. **AI-driven test generation strategy:** A workflow of draft generation, human review, traceability linkage back to requirements, and execution-metrics tracking works better than generating and trusting blindly. Feed the generator well-written requirements — poor requirements erode the speed gain under review. Track three sprint-level numbers: the percentage of generated tests that pass review without rework, the duplication rate against existing coverage, and acceptance-criteria coverage. In regulated domains, AI is a complement, not a replacement — exploratory testing and safety-critical assertions still need a human, with sign-off logged for audit purposes.
2. **Avoiding "test theater":** The core technique is mutation testing by hand: deliberately break the code under test and confirm the AI-generated test actually fails. Watch for vacuous assertions (checking that something merely exists rather than that it's correct) and for a second QA engineer to review test names, assertions, and data before trusting a generated suite.
3. **Visual regression testing with AI:** AI-driven UI changes introduce visual drift from several sources: model non-determinism, multi-file agentic rewrites that spread risk across a UI, design-token drift, and rendering issues invisible to text-based checks. Combine cheap pixel-diffing (fast, but prone to false positives from anti-aliasing or animation) with AI/perceptual diffing for structural-intent triage, and mask known-dynamic regions before layering AI triage on top — masking should come before intelligence, not after.
4. **Integrating AI QA into CI/CD at scale:** The same dual-path pattern from developer practices applies to QA gating: fast, non-production preview loops for AI-driven test generation and iteration, with unchanged strict gates — coverage thresholds, security scans, human approval — before anything reaches production.

> *"AI-generated assertions can confirm incorrect system behavior rather than detect it."*
> — Jama Software, on AI test case generation in regulated QA (2026)
>
> *"A test is not just a passing condition; it must assert the correct behavior."*
> — testRigor, on validating AI-generated tests (2026)
>
> *"AI-assisted UI development creates this gap: tools like Cursor, Copilot, and v0 produce code that clears text-based quality gates yet introduces spacing errors and design token drift visible only in browser rendering."*
> — Augment Code, on visual regression testing (2026)

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Mylar Technologies: In Practice

Mylar's QA lead adopted an AI test-generation tool the same week defect escapes from the new support tool had started climbing. The suite went green almost immediately — and that was exactly the moment to be suspicious of it, not reassured. A quick pass of deliberately broken test cases confirmed several generated tests were checking that the code ran at all, not that it returned the right answer.

When Things Don’t Go as Planned

When mutation testing or perceptual diffing tooling isn't available yet, fall back to the human checkpoints — a second QA engineer reviewing generated tests, manual sign-off in regulated domains — until the tooling catches up.

What You've Achieved

With this in place, your team has moved from generating AI tests to trusting them appropriately: every generated suite has been stress-tested against real failures, every visual diff has been triaged rather than rubber-stamped.

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Chapter 23: Agentic AI

Setting the Scene

Agentic AI dominates vendor pitches and board conversations in 2026, but the term is used so loosely that it now covers everything from a single chatbot plugin to a fully autonomous multi-step system. Before committing budget or headcount, it's worth understanding what the term is actually supposed to mean.

Where Most Teams Start

Without a working definition, most teams default to labeling anything with an LLM and a tool call "agentic," then either chase every vendor demo that uses the word or dismiss the category as hype altogether.

A Better Approach

The alternative is a disciplined, skeptical-by-default posture toward agentic claims — treat the label as a hypothesis to test against a specific, bounded process, not a feature to buy off a slide.

"Agentic AI" is one of the most overused terms in the current market, which makes a plain, careful definition worth having before deciding anything based on it.

1. **What it actually means:** Anthropic draws the clearest technical line: a workflow is an LLM-plus-tools system orchestrated through predefined code paths, while an agent is a system where the model dynamically directs its own process and tool use, maintaining control over how it accomplishes a task. McKinsey's more operational definition: a system based on generative AI foundation models that can act in the real world and execute multistep processes. The practical distinction from a chatbot is autonomy over multiple steps, not just a longer conversation.
2. **Where it's actually landing (2026):** Enterprise use cases with real traction include contract review with learning loops, financial-document extraction and validation, multi-step insurance claims investigation, and KYC/credit-risk analysis in banking. But adoption is far behind the marketing: one 2026 industry estimate put deployed (not piloted) usage at only about 17% of organizations, even though more than 60% expect to deploy within two years — a substantial intent-versus-readiness gap.
3. **Risks worth taking seriously:** Analyst research has been blunt: most agentic AI projects today are early-stage experiments driven by hype rather than clear value, and a widely cited 2025 prediction expects more than 40% of agentic AI projects to be canceled by the end of 2027 due to escalating costs and unclear ROI. Watch for "agent-washing" — vendors rebranding existing chatbots or RPA tools as agentic without real autonomous capability. McKinsey's own framing of the change-management burden is worth internalizing directly: onboarding an agent is closer to hiring a new employee than deploying a piece of software.

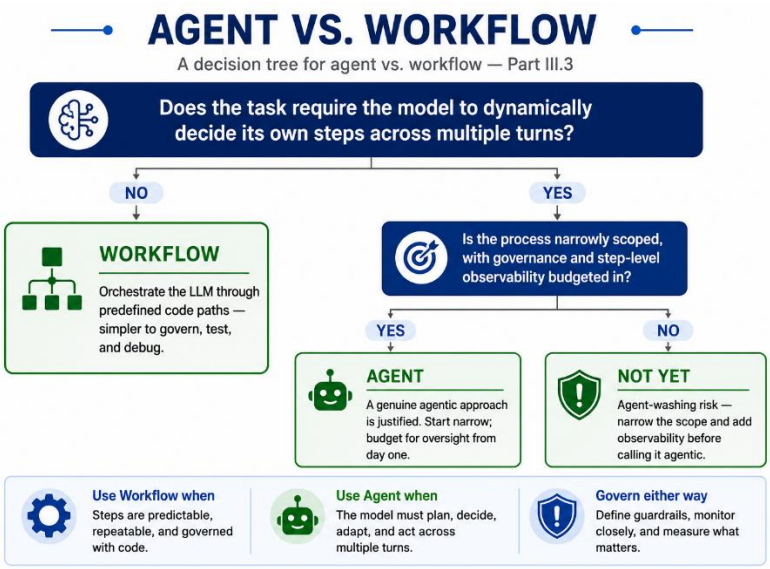
Mylar Technologies: In Practice

Mylar's support tool never became an agent, and that was a deliberate choice, not an oversight. A ticket-response draft follows the same steps every time: retrieve the account record, retrieve the relevant documentation, draft a reply, flag it for a human rep to send. That is a workflow, not an agent — and a workflow was the right call, because every step needed to be auditable and nothing about the task benefited from letting the model choose its own order of operations.

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The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.

Decision Tree — Agent vs. Workflow



Practical guidance

Treat agentic claims skeptically by default. Start narrowly scoped — one well-bounded process, not an open-ended mandate. Budget for governance, observability, and step-level tracking from day one, not as an afterthought once something goes wrong.

> *"Most agentic AI projects right now are early stage experiments or proof of concepts that are mostly driven by hype... Most agentic AI propositions lack significant value or return on investment."*
> — Anushree Verma, Senior Director Analyst, Gartner (2025)
>
> *"Onboarding agents is more like hiring a new employee versus deploying software."*
> — McKinsey QuantumBlack, "One year of agentic AI" (2025)
>
> *"Only 17% of organizations have deployed AI agents to date, yet more than 60% expect to do so within the next two years."*
> — Gartner, 2026 Hype Cycle for Agentic AI

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When Things Don’t Go as Planned

Not every process is ready for agentic treatment, and pushing ahead anyway is how the risks below materialize. If your organization lacks the observability, governance, or process maturity an agent needs, the right move is to scope back down to a workflow — or wait — rather than force autonomy prematurely.

What You've Achieved

Applied well, this chapter's distinctions turn a hyped, undifferentiated buzzword into an operational filter: you can now tell a real agent from a rebranded workflow, and set expectations that match the current state of the technology rather than its marketing.

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Chapter 24: RAG — Retrieval-Augmented Generation

Setting the Scene

Knowledge inside an organization changes constantly, while a language model's training data is frozen the moment training ends. Teams that ignore this gap ship AI systems that sound confident while being quietly out of date.

Where Most Teams Start

Most teams start by either prompting an off-the-shelf model and hoping it already knows enough, or manually pasting a few documents into the context window whenever an answer needs to be current. Neither approach scales.

A Better Approach

The shift RAG asks for is to stop treating the model's training data as the source of truth and instead connect the model to a live, external knowledge base that is queried at the moment of the question.

In plain terms: RAG connects a language model to your organization's own documents at the moment of the question, retrieving relevant passages before the model answers — the difference between an employee who looks up the current policy document versus one who answers from memory.

1. **When to use RAG:** When information changes frequently, answers must be traceable to a source, you don't have labeled training data, or you need fast deployment with fine-grained control over which sensitive data the model can see.
2. **When to use fine-tuning instead:** When you need a consistent domain-specific tone or output format, the underlying knowledge is stable rather than constantly changing, query volume is high enough to justify the training investment, or you need lower per-query latency than a retrieval step allows.
3. **When plain prompting is enough:** When an off-the-shelf model is already good enough for the task and no proprietary knowledge is required.
4. **The trade-offs, side by side:** RAG has no training cost but ongoing infrastructure (a vector database to maintain); fine-tuning has a high upfront cost but is cheaper and faster per query afterward; prompting is cheapest and fastest of all but weakest on proprietary or fast-changing knowledge. For high-value, high-volume use cases, a hybrid of fine-tuning and RAG together typically outperforms either alone.

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> *"Retrieval augmented generation connects large language models to an external knowledge base at inference time... this enables AI models to access current data without retraining."*
> — Databricks Blog, "RAG vs Fine Tuning" (2026)
>
> *"The hybrid approach — combining fine tuning with RAG retrieval — typically outperforms either method in isolation."*
> — Databricks Blog (2026)

Mylar Technologies: In Practice

Mylar's actual implementation retrieves two things at query time: the specific client's ticket history and the relevant product documentation page, both scoped to that one account so a rep at one client never sees another client's data. Getting the retrieval scope right, not the model choice, was most of the engineering effort — exactly matching the architecture decision made back in Chapter 6.

When Things Don't Go as Planned

If RAG's ongoing infrastructure overhead isn't justified for your case, or the knowledge domain is stable rather than constantly moving, the decision points below tell you where to go instead of forcing RAG onto a use case it doesn't fit.

What You've Achieved

With this discipline in place, your team no longer defaults to RAG, fine-tuning, or prompting out of habit — each is chosen deliberately based on how the knowledge behaves and what the use case demands.

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Chapter 25: MCP — Model Context Protocol

Setting the Scene

MCP earns a place in this playbook because of how fast it moved. A protocol few outside engineering had heard of two years ago is now a board-level topic — because it answers a problem every enterprise adopting AI agents runs into: connecting an assistant to the many systems it needs to be useful.

Where Most Teams Start

Most organizations arrive at that problem the same way: one integration at a time, hand-wired, with no shared pattern for access control between them.

A Better Approach

The shift this chapter argues for is protocol over patchwork: standardize the connection once, and use that standardization point to enforce accountability.

MCP is a standardized, open interface that lets AI agents and assistants connect to enterprise tools and data sources — databases, SaaS apps, internal systems — without a custom-built integration for every single one. One industry writer's shorthand: the USB-C of AI, a universal connector.

1. **Why it matters for enterprises:** Before MCP, connecting an AI agent to each internal system meant bespoke API glue code, repeated for every tool. MCP standardizes that connection once. Just as importantly for governance, it's designed to let agents act on behalf of users while preserving user-level context and accountability — a shift away from broad system-level credentials toward traceable, individually accountable operations.
2. **Where adoption stands in 2026:** MCP has gone from an obscure technical concept to an executive-level topic in roughly a year. It's now embedded in vendor enterprise products, widely used by coding assistants, and already operational in incident management, support-ticket prioritization, and workflow automation at a number of organizations.
3. **The risk to flag:** Security researchers have warned that MCP's default security controls are still thin relative to the speed of its adoption, and organizations should treat MCP servers as new attack surface requiring the same access-control rigor as any other integration point — not a shortcut around it.

> *"Instead of having an API tightly defined between a client and a server, you put an LLM on either end and let them negotiate what to exchange."*
> — Andy Ellis, security executive, via CIO.com (2026)

> *"The relative lack of default security controls, combined with the speed of adoption, means organizations should anticipate destructive incidents."*
> — Security researchers, RSA Conference 2026, via CIO.com

Mylar Technologies: In Practice

Mylar's support tool eventually needed to read from two systems that were never designed to talk to each other: the ticketing platform and the internal documentation wiki. Rather than writing a custom integration for each new tool the assistant might need, Mylar connected both through MCP, so adding a third data source later — the billing system, say — would mean writing one new connector, not rewiring the assistant itself.

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When Things Don’t Go as Planned

If your security team can't yet vouch for an MCP server's access controls, that's a stop sign, not a footnote — revisit before granting it production data access. Treat every MCP connection as new attack surface, governed like any other integration point.

What You've Achieved

Once this is in place, MCP stops being a novelty and becomes infrastructure: one standardized, auditable way for agents to reach enterprise systems, with the security discipline to match its now-executive-level visibility.

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Chapter 26: Local vs. Cloud Models

Setting the Scene

By 2026, most organizations running AI in production hit the same fork: should this workload run on infrastructure you control, or on a vendor's hosted API? The answer shapes cost, compliance exposure, and operational burden.

Where Most Teams Start

Most teams don't consciously choose between local and cloud models. They default to whichever cloud API got a demo working fastest, or they mandate "local only" on general security instinct, without checking whether local hardware can actually run a model capable enough for the job.

This is fundamentally a "build vs. rent" decision: run models on your own hardware using open-weight models, versus calling a vendor's hosted API over the internet.

1. **Local makes sense when:** Data sovereignty or offline operation genuinely matters, or you're running high enough volume that a fixed hardware cost beats ongoing per-call pricing. The trade-off is that local hardware caps how large and capable a model you can practically run, and you take on the maintenance and security burden yourself.
2. **Cloud makes sense when:** You need frontier-level capability, elastic scaling without buying hardware, and want the vendor to own maintenance and security at their scale. The trade-off is ongoing usage-based cost, network latency, and data leaving your premises — which has to be mapped against your own compliance requirements.
3. **The 2026 practical answer for most organizations:** A hybrid split: smaller, local or open-weight models for routine, sensitive, or very high-volume tasks, and cloud-hosted frontier models for complex or lower-volume, higher-value tasks. This pattern shows up repeatedly across independent 2026 vendor and analyst comparisons.

Mylar Technologies: In Practice

Local models were never seriously on the table for Mylar. At 45 people with no dedicated infrastructure team, the operational cost of hosting and maintaining a model outweighed any latency or data-residency benefit, and none of the client contracts required on-premises processing. The pilot ran entirely on a hosted cloud model from day one — the practical default for most organizations this book is written for.

> *The formal method behind this follows below — useful if you want to run it yourself, optional if the takeaway above is all you needed.*

Decision Tree — Build vs. Rent (Hybrid Split)

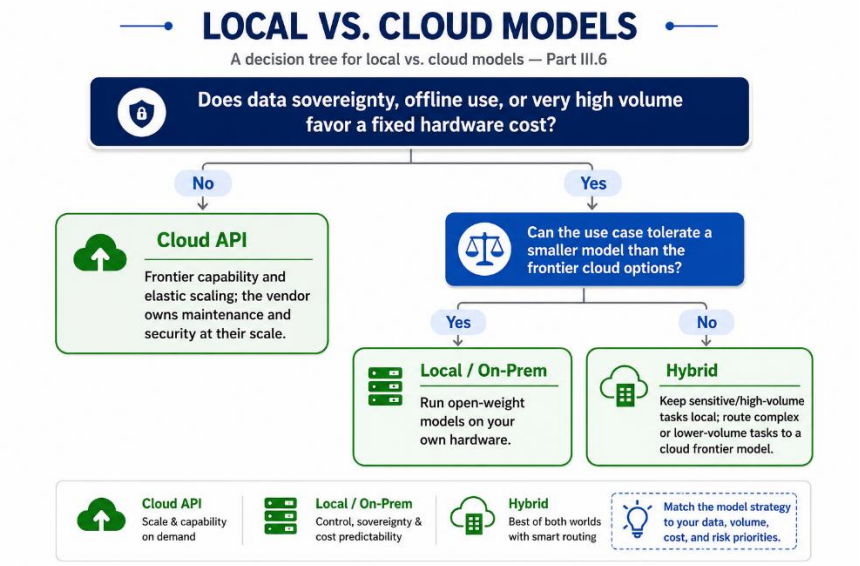
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Local vs. Cloud AI Models

A build-vs-rent decision, factor by factor (Section III.6)

Factor	Local / On-Prem	Cloud API
Data privacy	Data stays on-device; you own security	Provider secures at scale; data leaves the premises
Cost model	One-time hardware investment	Pay-as-you-go, accumulates with usage
Latency	Lower — no network hop	Higher — network round-trip
Capability	Capped by local hardware (smaller models)	Access to the largest frontier models
Maintenance	You maintain updates and patching	Provider handles maintenance
Scalability	Requires physical hardware upgrades	Elastic, on-demand scaling
Connectivity	Works offline	Requires stable internet

Decision Tree — Local vs. Cloud Models



> *"Running a model locally can reduce latency since data does not need to be sent over the network... local devices may have limitations on the size and complexity of models that can be run due to hardware constraints."*
> — Microsoft Learn, "Choose between cloud-based and local AI models" (2026)
>
> *"Cloud platforms offer easy scalability, allowing you to quickly adjust resources based on demand without the need for physical hardware changes."*
> — Microsoft Learn (2026)

When Things Don’t Go as Planned

If a workload doesn't cleanly fit either bucket, don't force a permanent choice. Default to cloud for capability gaps or local for sovereignty gaps as a stopgap, and revisit the split as hardware, pricing, or vendor terms change.

What You've Achieved

Once this split is in place, your team stops re-litigating the local-vs-cloud question for every new use case — it applies a known framework, checks it against the decision tree, and moves on.

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Chapter 27: ROI Measurement

Where Most Teams Start

Most teams start by counting what's easy: how many people logged in, how many suggestions got accepted, how much faster a task felt. Those numbers get reported up the chain as "ROI," often echoing vendor claims of 50-100% productivity gains, with no baseline and no accounting for human review time.

A Better Approach

A better approach treats ROI as a case you build, not a number you assert. That means anchoring every claim in a defined baseline and a small, consistent set of dimensions.

Measuring AI ROI in practice is harder than the marketing suggests, and getting it wrong in either direction — overclaiming or under-measuring — undermines the whole program's credibility with leadership.

1. **A workable three-dimension frame:** Utilization (is it actually being used, and how often), Impact (productivity, quality, developer/employee experience), and Cost (total spend against total return). One large multi-company study of nearly 39,000 developers found real productivity gains landing at 5–15% — not the 50–100% figures often marketed — once measured this way.
2. **The single biggest measurement pitfall:** Human oversight and review time is routinely left out of the savings calculation. If a task takes an AI two minutes but a human ten minutes to review and correct, the real time saved is not "the AI did it in two minutes" — accepted-code or accepted-output rates are similarly misleading, since accepted output is frequently modified before it's actually used.
3. **Other common pitfalls:** Attribution confusion in mixed human-plus-AI workflows (tag each step as machine-generated, human-verified, or human-enhanced rather than crediting the outcome wholesale to AI); measuring model accuracy instead of what actually changed in the business; hidden total cost of ownership (integration, monitoring, and change management routinely left off the cost side); and assuming a pilot's ROI will simply scale — one large longitudinal study found some AI-heavy portfolios showing plateaued or even declining process effectiveness despite continued investment.
4. **A useful discipline: risk-adjusted ROI:** Discount the gross expected benefit by your observed hallucination rate, guardrail-intervention rate, and override rate. As a rule of thumb, target a payback period under two quarters for operational use cases and under a year for developer tooling.



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> *“Without baselines, you'll never know if AI improved productivity or just changed how work feels.”*
> — Taylor Bruneaux, DX (getdx.com), AI Measurement Framework (2025)
>
> *“Faster doesn't mean ROI.”*
> — Agustina Branz, Source86, via CIO.com (2025)

Mylar Technologies: In Practice

Mylar's actual ROI case, six months in, looked less impressive than the pitch had promised, and more trustworthy because of it. Raw time saved per ticket was real, but the risk-adjusted figure discounted it for the rework rate: replies a human rep had to substantially rewrite before sending. That discounted number, not the headline figure, is what let the CEO tell the board honestly that the pilot had paid for itself in under five months.

When Things Don’t Go as Planned

Measurement breaks down in predictable ways — know them going in, so a bad number doesn't kill a program that's actually working, and a good-looking one doesn't hide a program that isn't.

What You've Achieved

Once this discipline is in place, you're no longer reporting adoption or sentiment as if it were value — you're reporting a risk-adjusted return leadership can hold you to, quarter over quarter.

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Chapter 28: Enterprise Case Studies

Setting the Scene

Every framework and governance model in this playbook eventually faces the same question: does it hold up in production, at real companies, under real numbers? This chapter answers with six named enterprises rather than anonymized composites.

Where Most Teams Start

The common shortcut is to treat any "case study" as proof: lift the headline number from a vendor blog post and drop it into a deck, without asking who measured it, how, or whether anyone independent checked it.

Six real, named examples — each labeled by how independently verified the claim is, since a vendor's case study about its own customer and a peer-reviewed study are not the same kind of evidence, even when both are true.

Klarna — Agentic AI in production (customer service)

An OpenAI-powered assistant handling full conversational resolution across 23 markets and 35+ languages since February 2024. 2.3 million conversations in the first month alone (roughly 700 full-time-equivalent workers' worth of volume), average resolution time down from 11 minutes to under 2, and a projected $40 million 2024 profit impact; cost per transaction later fell 40%. Notably, in May 2025 Klarna's CEO publicly said the AI-first approach "went too far" on quality and the company began rehiring humans — an honest walk-back worth including alongside the headline numbers.

> **Credibility:** *Company primary source, corroborated by independent press (Bloomberg, Forbes) — including the walk-back.*

LinkedIn — RAG for internal customer-service resolution

A knowledge-graph-plus-LLM RAG system, evaluated through an actual randomized controlled trial: median resolution time down 28.6% (from 7 hours to 5).

> **Credibility:** *Independent, peer-reviewed (SIGIR 2024 paper) — the strongest evidentiary standard in this list.*

GitHub Copilot at Accenture — Developer practice transformation

A randomized controlled trial across thousands of developers found +8.69% pull requests per developer, +15% merge rate, +84% successful builds, and 90% of developers reporting they felt more fulfilled. Worth reading alongside two independent counterweights: METR's RCT found developers were actually 19% slower with AI tools despite believing they were faster, and Google Cloud's 2025 DORA report (roughly 5,000 respondents) found AI adoption boosts throughput but hurts stability.

> **Credibility:** *The Accenture/GitHub figures are a vendor case study (GitHub/Microsoft about their own customer) — present alongside the independent METR and DORA findings, not instead of them.*

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Meta — AI-generated testing (QA transformation)

LLM-based, mutation-guided test generation across seven Meta platforms including Messenger and WhatsApp, with a 73% engineer acceptance rate for AI-generated tests across nearly 11,000 Kotlin classes.

Credibility: *Independent, peer-reviewed (ACM FSE 2025 paper).*

DBS Bank — AI governance framework rollout

A published "PURE" governance framework (Purposeful, Unsurprising, Respectful, Explainable) with a senior oversight committee and mandatory human-in-the-loop review for sensitive functions. As of mid-2025: over 1,500 models and 370 use cases in production, with time-to-market for a new model cut from 15 months to under 3.

Credibility: *Company primary source, cross-referenced by Singapore's financial regulator (MAS).*

IBM — Reported ROI from internal AI automation

Internal AskHR/AskIT agentic tools deployed across more than 70 internal functions, with a claimed $3.5 billion productivity improvement over two years and 94% of routine HR tasks automated through AskHR.

Credibility: *Company claim, unaudited, no methodology disclosed — treat as directional, not verified.*

When Things Don’t Go as Planned

Two cases model the exception-handling worth internalizing here. Klarna's own CEO said its AI-first approach "went too far" and the company began rehiring humans — read that correction alongside the launch numbers, not instead of them. IBM's figures, labeled unaudited with no disclosed methodology, are this chapter's example of a claim to treat as directional and revisit before it carries a decision.

What You've Achieved

You now have six concrete reference points for AI in production, each tagged with how much independent scrutiny its numbers received. A number paired with a rated confidence in that number is what makes a business case defensible. Hold every future AI claim you meet to that same standard.

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Templates & Worksheets

The frameworks in this book only become a habit once they're easy to reuse. The eight worksheets below are ready to copy into a document or spreadsheet as-is — each mirrors a framework introduced earlier in the book, reduced to a fillable form.

1. AI Readiness Worksheet

1. **Based on:** Chapter 2

Dimension	Score (1-5)	Evidence / Notes
Leadership & Strategy		
Data Readiness		
Talent & Skills		
Infrastructure		
Process Maturity		
Governance		
TOTAL (max 30)		

2. AI Opportunity Canvas

1. **Based on:** Chapter 3

Field	Your Entry
Problem statement	
Current process (as-is)	
Data available	
AI capability match	
Expected value	
Risks	
Go / No-Go decision	

3. Business Case Template

1. **Based on:** Chapter 5

Field	Your Entry
Problem statement	
Expected financial impact	
Strategic outcomes	

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Field	Your Entry
Operational KPIs affected	
Adoption & engagement plan	
Technical performance threshold	
Total cost — build	
Total cost — run	
Risk assessment (NIST categories)	
Payback period estimate	
Decision gate date	

4. Architecture Decision Record

1. **Based on:** Chapter 6

Field	Your Entry
Decision title	
Context	
Build / Buy / Blend choice	
Model / provider choice & rationale	
Prompting / RAG / Fine-tuning choice & rationale	
Alternatives considered	
Consequences	
Owner & date	

5. Governance Checklist

1. **Based on:** Chapter 10

Item	Done? (Y/N)	Owner
AI lead named		
Written AI dos-and-don'ts policy		
Approved tool allow-list published		
Fairness & bias review (if applicable)		
Reliability & validity review process		

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Item	Done? (Y/N)	Owner
Privacy & security data rules defined		
Transparency standard defined		
Accountability owner named per decision type		
Legal / regulatory compliance reviewed		
Vendor / third-party risk reviewed		
Acceptable IP-use policy defined		
Quarterly policy review scheduled		

6. Quarterly Review Template

1. **Based on:** Chapter 17

Field	Your Entry
Level under review	
Current maturity stage (1-5)	
Target stage	
Primary KPI & current value	
Milestone checklist status	
Blockers	
Actions for next quarter	
Reviewer & date	

7. ROI Calculator

1. **Based on:** Chapter 27

Field	Your Entry
Utilization metric (how often actually used)	
Impact metric — before	
Impact metric — after	
Total cost — build + run + human review time	
Gross expected benefit	

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Field	Your Entry
Risk adjustment (hallucination / override rate discount)	
Net risk-adjusted ROI	
Payback period	

8. Risk Register

1. **Based on:** Chapter 5 / Chapter 10

Risk	Likelihood (L/M/H)	Impact (L/M/H)	Mitigation	Owner	Status
Data privacy exposure					
Hallucinated or fabricated output					
Vendor / third-party lock-in					
Bias in AI-assisted decisions					
IP ownership / attribution ambiguity					
(add your own)					
(add your own)					

CHAPTER RECAP

1. Eight worksheets turn this book's frameworks into fillable forms: Readiness, Opportunity Canvas, Business Case, Architecture Decision Record, Governance Checklist, Quarterly Review, ROI Calculator, and Risk Register.
2. Copy any of them into a document or spreadsheet as-is — each mirrors a framework introduced earlier in the book.

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Future Roadmap: AI Evolution 2026–2030

Everything before this chapter is written to be actionable today. This chapter is different in kind: it's a deliberately hedged look at where the current signals point over the next several years, so a transformation built on this book's framework doesn't get blindsided by a shift its own foundation should have anticipated. Treat every date below as directional, not a delivery commitment — the analysts cited here have already had to revise similar predictions once.

2026 — Agent proliferation meets a maturity shakeout, simultaneously

Both things are true at once this year, and that tension is the defining feature of the period. On one hand, task-specific AI agents are being embedded into enterprise applications fast — Gartner projects roughly 40% of enterprise applications will feature task-specific AI agents by 2026, up from under 5% in 2025. On the other hand, the same analyst house has been blunt that most of what's shipping under the "agentic" label is still early-stage experimentation: Gartner's own prediction is that more than 40% of agentic AI projects will be canceled by the end of 2027, over escalating cost and unclear ROI. For an organization using this book, that means Part III.3's guidance — start narrow, budget for governance and observability from day one — is not a conservative hedge, it's the specific practice that determines which side of that 40% cancellation line an initiative lands on.

2027 — The cancellation wave, and what survives it

The projects that get canceled by end of 2027 will disproportionately be the ones that skipped the discipline this book describes: no readiness assessment (Chapter 2), no governance foundation (Chapter 10), agentic scope that was never narrowed (Part III.3), and ROI measurement that never accounted for human review time (Chapter 27). The projects that survive will look, by definition, like the ones built the way this book recommends. This is also the period analysts expect enterprise AI spending to shift from broad experimentation toward a smaller number of well-governed, well-measured initiatives — consolidation, not retreat.

2028 — Agents as economic actors, not just assistants

Gartner has projected that AI agents will directly influence roughly $15 trillion in B2B purchasing activity by 2028 — agents that don't just draft a recommendation for a human buyer, but participate directly in sourcing, negotiation, and procurement workflows. That shift raises the stakes on exactly the governance question this book treats as foundational: a named human is always accountable for a decision (Principle 4), and that principle gets harder, not easier, to uphold as agents take on economically consequential actions with less direct human involvement at each step.

2029-2030 — Agents, robots, and people become a designed partnership

McKinsey's research on "agents, robots, and us" frames the emerging model directly: the future of work is a deliberate three-way skill partnership between people, software agents, and physical robots, not a simple substitution of one for the other. Reasoning models continue to mature past pattern-matching toward genuine multi-step problem decomposition, and "physical AI" — AI systems that perceive and act in physical space — moves from warehouses and supply chains (an area Gartner already flags as a near-term physical-AI entry point) toward broader operational use. The organizations best positioned for this shift will be the ones that treated AI adoption as an operating-model question all along — which is the entire premise of this book's Role Responsibility Matrix and six-level structure, not a bolt-on afterthought once robots show up.

None of this changes what to do this quarter. It changes what to watch for: if agent-driven purchasing, physical AI, or reasoning-model capability jumps arrive faster or slower than expected, the response is the same either way — revisit the readiness assessment, tighten the governance foundation, and let the numbers from Chapter 27 decide whether to scale, not the headline.

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> *“More than 40% of agentic AI projects will be canceled by the end of 2027, due to escalating costs, unclear business value or inadequate risk controls.”*
> — Gartner, press release (2025)
>
> *“40% of enterprise applications will feature task-specific AI agents by the end of 2026, up from less than 5% in 2025.”*
> — Gartner, press release (2025)
>
> *“The future of work is not people versus machines, but people, agents, and robots learning to work together.”*
> — Paraphrased summary of McKinsey Global Institute's framing, "Agents, robots, and us: Skill partnerships in the age of AI" (2026)

CHAPTER RECAP

1. 2026 brings simultaneous agent proliferation and a maturity shakeout — both are true at once, and this book's discipline determines which side of the shakeout an initiative lands on.
2. 2027's cancellation wave will disproportionately hit projects that skipped readiness, governance, narrow agentic scoping, or honest ROI measurement.
3. By 2028, Gartner projects agents will influence roughly $15 trillion in B2B purchasing activity — raising the stakes on the “a human is always accountable” principle.
4. 2029–2030: McKinsey frames the endpoint as a three-way skill partnership between people, agents, and robots — not a simple substitution.

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Conclusion & Summary

Three parts, one methodology. Part I made sure the organization spends its first dollar of AI investment on a real, prioritized opportunity rather than a trending one. Part II made adoption measurable and consistent across six organizational levels instead of leaving it to accumulate unevenly and invisibly. Part III made sure that once adoption is underway, the technical execution — the code, the tests, the agents, the architecture — holds up to real scrutiny rather than marketing claims.

The throughline across all three parts is the same discipline applied at a different altitude: be honest about where you actually stand before deciding what to do next. A readiness score that hasn't been measured, a maturity stage that hasn't been diagnosed, and an ROI figure that hasn't accounted for human review time are the same mistake in three different places — optimism substituting for measurement. Every framework, matrix, and diagnostic question in this book exists to close that gap.

This is a living methodology, not a one-time audit. Revisit the readiness scorecard (Chapter 2) and the maturity scorecard (Chapter 17) on the same quarterly cadence recommended in Section 5 — capability changes, tools change, and the organization's own stage will move. What should not change quarter to quarter is the discipline itself: assess honestly, prioritize deliberately, govern lightly but firmly, execute with the failure modes in Part III in mind, and measure the result in a way that would survive being shown to a skeptic.

Where to go next depends on where the organization actually is. An organization that has not yet run a readiness assessment should start at Chapter 2, today. An organization mid-rollout with no governance foundation should stop and read Chapter 10 before doing anything else. An organization already deploying agentic tooling without the practices in Chapter 21–III.3 is carrying more risk than it has priced in. The Appendix that follows is built to make it fast to find the exact tool needed for wherever that starting point turns out to be.

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Appendix — Frameworks, Matrices, and Scorecards at a Glance

Every framework, matrix, scorecard, and checklist introduced across the three parts is indexed here by name, purpose, and location — a fast way back to a specific tool without re-reading the chapter that introduced it.

Framework / Tool	Purpose	Section
AI Readiness Scorecard	Six-dimension 1–5 self-assessment (Leadership, Data, Talent, Infrastructure, Process, Governance) producing a readiness tier.	Part I.1
AI Opportunity Canvas	Structured ~45-minute worksheet per candidate process: problem, data, capability match, value, risk, go/no-go.	Part I.2
Pain Point × Pressure Matrix	Scores candidate pain points 1–10 on severity and urgency before committing to a full canvas.	Part I.2
Value Stream Mapping	End-to-end process mapping to locate where waste and delay actually accumulate.	Part I.2
Prioritization Matrix (Value vs. Effort)	Four-quadrant plot — Quick Wins, Strategic Bets, Minor Automations, Distractions — for ranking opportunities.	Part I.3
RICE-A Scoring	Reach × Impact × Confidence ÷ Effort, with an added AI Ambiguity discount factor.	Part I.3
Business Case Template (Five-Layer + NIST Risk)	Financial, strategic, operational, adoption, and technical benefit layers paired with a structured AI risk assessment.	Part I.4
AI Architecture Decision Record	One-page record of build/buy/blend mode, model/provider rationale, and prompting/RAG/fine-tuning choice.	Part I.5
Prompting vs. RAG vs. Fine-Tuning Decision Guide	Side-by-side cost, latency, knowledge-freshness, and maintenance comparison.	Part I.5 / III.4
Universal AI Adoption Maturity Ladder	The shared five-stage scale (Aware → Experimenting → Adopting → Integrated → Optimizing) used by every level.	Part II, Sec. 2
Governance Pillars Checklist	Nine-pillar lightweight governance checklist (fairness, reliability, privacy, transparency, accountability, and more).	Part II, Sec. 3
150 Stage Diagnostic Questions	Five diagnostic questions per stage, per level (6 levels × 5 stages), each with an example and explanation.	Part II, Sec. 4
Per-Level Milestone Checklists and KPIs	Qualitative checklist plus quantitative KPI set for each of the six organizational levels.	Part II, Sec. 4
Maturity Scorecard	Single-page 1–5 rating of all six levels against current and target stage, with primary KPI to watch.	Part II, Sec. 5
Suggested Rollout Timeline	A 0–12+ month phased rollout sequence across the six levels.	Part II, Sec. 6
AI Failure-Mode Checklist (Developer)	Eight named failure modes — Hallucination Spiral, Comprehension Debt, Automation Bias, and more.	Part III.1
Dual-Path CI/CD Pattern	Fast non-production experimentation path paired with an unchanged strict production gate.	Part III.1 / III.2

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Framework / Tool	Purpose	Section
Test-Theater Detection Technique	Manual mutation testing to confirm AI-generated tests actually fail on broken code.	Part III.2
Agentic AI Risk Guidance	Narrow-scope-first guidance plus the workflow-vs-agent technical distinction.	Part III.3
Local vs. Cloud Decision Table	Seven-factor build-vs-rent comparison: privacy, cost, latency, capability, maintenance, scalability, connectivity.	Part III.6
Three-Dimension ROI Frame (Utilization / Impact / Cost)	The measurement frame plus the risk-adjusted ROI discipline for discounting gross benefit by observed error rates.	Part III.7
Case Study Credibility Framework	Labels every case study as vendor-published, company self-reported, or independent/peer-reviewed.	Part III.8
AI Transformation Journey	Nine-stage end-to-end lifecycle (Discover → Institutionalize), mapped to Parts I–III.	Journey chapter
Ten AI Transformation Principles	Agile-Manifesto-style memorable set, each cross-referenced to a specific chapter.	Principles chapter
Top AI Transformation Mistakes	Eight organizational/strategic anti-patterns, each with a named fix.	Anti-patterns chapter
Five Decision Trees	Build vs. Buy, Prompting vs. RAG vs. Fine-Tuning, Local vs. Cloud, Agent vs. Workflow, Pilot vs. Production.	I.5 / III.3 / III.4 / III.6 / II §3
AI Operating Model & RACI Matrix	Four-tier org chart plus a 10-decision Responsible/Accountable/Consulted/Informed matrix across 8 roles.	Part II §3
Capability Matrix Across Organization Sizes	Illustrative maturity target, investment, ROI timeframe, governance effort, and complexity by org size.	Capability Matrix chapter
Templates & Worksheets (8)	Fillable native tables: Readiness, Opportunity Canvas, Business Case, ADR, Governance Checklist, Quarterly Review, ROI Calculator, Risk Register.	Templates chapter
Future Roadmap 2026–2030	Four analyst-grounded horizons on agents, physical AI, and reasoning models, explicitly hedged as directional.	Roadmap chapter

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Applying the Framework: Six Standpoints at Mylar Technologies

Every framework in this book has appeared one piece at a time — a scorecard here, a decision tree there, a maturity stage worked through on its own. Before closing, it's worth seeing what it looks like assembled: one organization, all six levels, each standpoint applying the same discipline to its own slice of the problem. This is the same Mylar Technologies whose support-ticket pilot opened the Executive Summary and ran through Parts I and III — a fictional small scale software company building scheduling and workflow integrations for mid-market clients, not a verified case study like the six in Chapter 28, but a composite walkthrough showing how the book's own tools connect in practice.

Enterprise Standpoint

Where the mandate starts

Mylar's leadership team had been talking about AI for two quarters — a Slack channel, a few personal ChatGPT habits, no real decision. The CEO finally ran the AI Readiness Scorecard from Chapter 2 with her three direct reports in the room. The result stung a little: a 14 out of 30, dragged down by a 1 on Governance and a 2 on Data Readiness. Rather than treat that as a verdict, the leadership team treated it as a map — naming an AI lead that same week, publishing a one-page dos-and-don'ts policy before any tool was purchased, and setting a single, honest rule: no department gets a budget line for AI until the six-dimension score says the organization can actually support it.

The takeaway: a low score isn't a failure state, it's the first real data point Mylar had about itself.

Management Standpoint

Where policy becomes practice

Priya, who manages Mylar's customer success team, got the new AI policy in her inbox like everyone else. What made it stick wasn't the memo — it was the AI Operating Model & Role Responsibility Matrix from Chapter 10, which named her specifically as Accountable for any AI tool her team adopted. When a rep proposed an AI drafting tool for renewal emails, Priya didn't just approve it; she walked the proposal through the RACI matrix, named herself as the reviewer of first output, and set a 30-day check-in before letting it become the default.

The takeaway: a manager named in a matrix behaves differently than a manager cc'd on a memo.

Team Standpoint

Where a habit becomes an asset

One engineer on Mylar's platform team had been quietly using an AI assistant to draft release notes for months — good notes, but only when he remembered to write them and only in his own voice. When a teammate covered for him during a leave, the release notes reverted to terse one-liners, and nobody had a shared prompt or workflow to fall back on. The team's fix, straight out of Part II's team-level guidance: turn the habit into a shared doc — the actual prompt template, the review step, and a rule that any team member could run it, not just its inventor.

The takeaway: a practice that only survives one person's presence isn't a team capability yet.

Project Standpoint

Where disclosure meets a client

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A mid-sized client project — a custom scheduling integration — had quietly used an AI coding assistant for about a third of its codebase by the time the client's own security review asked, directly, "was any of this AI-generated, and who checked it?" Mylar's project lead didn't have a ready answer, and improvised one under pressure. The fix adopted afterward: every new project deliverable now carries the lightweight AI-disclosure note from the Project Level's milestone checklist, filled in as the work happens rather than reconstructed after the fact.

The takeaway: the right time to be able to answer that question is before it's asked, not after.

Developer Standpoint

Where shadow use becomes governed practice

Three of Mylar's eight developers were already using a personal AI coding subscription before the company had a policy — unofficial, untracked, and invisible to code review. Once engineering leadership approved a single tool for the whole team, the real change wasn't the tool itself; it was the one-line addition to the pull request template: "Contains AI-assisted code — reviewed to the same standard." Cycle time per PR dropped from just under two days to a day and a half within a quarter, and it was visible in the tracker, not just felt.

The takeaway: the productivity gain was already happening — governance just made it something the team could see and trust.

QA Standpoint

Where a passing suite earns its trust

Mylar's QA lead adopted an AI test-generation tool the same week defect escapes had started climbing. The suite went green almost immediately — and that was the problem. A quick mutation-testing pass (deliberately breaking three functions the new tests were supposed to cover) showed two of them still passed on broken code. Rather than rip the tool out, the team added the one missing discipline: a second engineer spot-checking generated assertions before trusting a suite, exactly as Chapter 22 describes. Within two months, test-suite maintenance time was tracked, trending down, and defect escapes were falling for the first time in a quarter.

The takeaway: a green checkmark is a claim, not proof — and it's worth five minutes to confirm it.

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Glossary

Key terms used throughout this book, gathered in one place for quick reference. Each entry points back to the section where the concept is introduced in full.

Agentic AI

A system where the model dynamically directs its own steps and tool use across a multistep task, as opposed to a workflow, where an LLM is orchestrated through predefined code paths. See Part III.3.

AI Readiness Scorecard

The six-dimension (Leadership, Data, Talent, Infrastructure, Process, Governance) 1-5 self-assessment used to establish a readiness tier before committing budget to a pilot. See Chapter 2.

BYOAI

"Bring your own AI" — individual or team adoption of an AI tool without a company-sanctioned policy or procurement process, the seed of Shadow AI.

Build vs. Buy vs. Blend

The first of three linked architecture decisions in Chapter 6: whether an AI capability is embedded in existing software, bought as a point solution, or built and blended in-house.

Fine-tuning

Additional training that adapts a model's weights to a specific domain, tone, or behavior — contrasted with RAG and prompting in Chapter 6 / III.4.

Governance Pillars

The nine-pillar lightweight governance checklist (fairness, reliability, privacy, transparency, accountability, and more) applied enterprise-wide. See Chapter 10.

Hallucination

A model output that is fabricated or factually wrong while still appearing fluent and confident — the central risk that risk-adjusted ROI and human review practices in this book are built to catch.

Maturity Ladder / Maturity Stage

The shared five-stage scale (Aware, Experimenting, Adopting, Integrated, Optimizing) used to diagnose every one of the six organizational levels. See Part II, Section 2.

MCP (Model Context Protocol)

A standardized, open interface that lets AI agents and assistants connect to enterprise tools and data sources without a bespoke integration for each one. See Part III.5.

Pain Point

A specific, named process failure — not a vague frustration — used as the starting unit of analysis in opportunity identification. See Chapter 3.

RACI

Responsible, Accountable, Consulted, Informed — the role-assignment framework used in the AI Operating Model & Role Responsibility Matrix. See Chapter 10.

RAG (Retrieval-Augmented Generation)

An architecture that retrieves relevant passages from an organization's own documents at the moment of a question, before the model answers, rather than relying solely on training data. See Chapter 24.

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Risk-Adjusted ROI

An AI return-on-investment figure discounted by the observed hallucination rate, guardrail-intervention rate, and override rate — the discipline recommended in Chapter 27 to keep ROI claims defensible.

RICE-A Scoring

Reach x Impact x Confidence / Effort, with an added AI Ambiguity discount factor, used to rank candidate opportunities. See Chapter 4.

Shadow AI

Employees adopting AI tools on their own, unofficially, because no sanctioned option exists — the first of the anti-patterns catalogued in Top AI Transformation Mistakes.

Test Theater

AI-generated tests that pass in CI without actually testing anything real — detected by deliberately breaking the code under test and confirming the test fails. See Chapter 22.

Value Stream Mapping

End-to-end mapping of a process as it actually happens, used to locate where waste and delay accumulate before applying AI to it. See Chapter 3.

Vendor Case Study vs. Peer-Reviewed Study

A distinction this book insists on throughout Chapter 28: a vendor's published account of its own customer is not the same evidentiary standard as an independently conducted or peer-reviewed study, even when both describe the same real deployment.

Workflow

An LLM-plus-tools system orchestrated through predefined code paths, as distinct from an agent, which dynamically directs its own process. See the Agent vs. Workflow decision tree in Part III.3.

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Note: "Mylar Technologies" is an illustrative composite created for this document, not a published case study.

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Thank you for reading — Hemoji Rao G, Independent AI Practitioner

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